The Effect of Emergency Financial Assistance on Employment and Earnings

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Abstract

We examine the labor supply effects of short-term income transfers for families experiencing a housing crisis. We link callers to an emergency assistance homelessness prevention hotline to their federal tax records and measure their earnings in years surrounding their calls. Our methodology exploits quasi-random variation in the availability of assistance to compare similar families receiving and not receiving funds. Looking up to four years post-assistance, we find evidence, especially for the lowest earners, of earnings gains, and overall we find no evidence that assistance lowers earnings. Our results indicate that any income effect of temporary transfers for those in crisis is minimal and that these targeted transfers may convey labor market benefits for the poorest of the poor.

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1. Introduction

A primary goal of social safety net programs is to help households maintain economic well-being when faced with a transitory shock (Gruber, 1997, 2000). Research has shown that such shocks, particularly for the poor, can create a downward spiral of economic challenges. For example, a temporary loss of income may result in an inability to pay rent and ultimately eviction, which, in turn, leads to increased likelihood of homelessness and reduced earnings (Collinson, Humphries, Mader, Reed, Tannenbaum, and Van Dijk, 2024). To prevent this type of chain reaction, most US communities have programs that make emergency financial assistance available to low-income families facing housing instability. This type of aid expanded dramatically during the COVID-19 pandemic, with Congress allocating more than $70 billion to state and local governments for emergency housing assistance (CARES Act, 2020; Consolidated Appropriations Act, 2020; American Rescue Plan Act, 2021).

However, there is little direct evidence on whether such assistance can preserve or improve earnings and employment for those experiencing a housing crisis. On the one hand, an income transfer that targets those in crisis may improve a recipient's economic stability, leading to better job search or job matching outcomes and increased employment or earnings (Desmond, 2016; Collinson et al., 2024). Moreover, policymakers often claim that housing assistance stabilizes recipients' lives in a way that better positions them to work. For example, the executive director of the US Interagency Council on Homelessness stated, "...[H]ousing is indeed the foundation for getting and keeping a job, addressing mental health and substance use, and reconnecting with family, friends, and community" (Olivet, 2023). A growing body of research indicates that temporary income transfers targeted towards those facing a housing crisis can have important nonpecuniary benefits for recipients, including reduced homelessness (Evans, Sullivan, and Wallskog, 2016; Phillips and Sullivan, 2025) and arrests for violent crimes (Palmer, Phillips, and Sullivan, 2019).

On the other hand, behavioral responses to short-term income transfers might lessen the incentive to work. A simple labor supply model predicts that a pure non-labor income

transfer such as emergency financial assistance would reduce labor supply if leisure is a normal good. A large literature has shown evidence of a negative income effect in a variety of contexts, including the Negative Income Tax experiments (Robins, 1985; Price and Song, forthcoming) and cash transfers that result from lotteries (Imbens, Rubin, and Sacerdote, 2001; Cesarini, Lindqvist, Notowidigdo, and Östling, 2017; Golosov, Graber, Mogstad, and Novgorodsky, 2024), but these studies do not focus on transfers that target disadvantaged groups such as those at risk of eviction. As we discuss below, work on more permanent housing-based transfers for the disadvantaged has found both negative and positive effects on employment and earnings, although it is unclear if these results would extend to a temporary emergency assistance context.

We examine whether short-term income transfers for housing affect labor supply using a quasi-experimental setting where, conditional on a set of observable characteristics, the availability of funding is exogenous. In particular, we examine the effect of emergency financial assistance coordinated by the Homelessness Prevention Call Center (HPCC) in Chicago, Illinois, which connects individuals and families in need with agencies that provide one-time assistance with rent (about $900-1,000) or utilities.

Availability of funding varies over time and across agencies and is, at any moment, unobservable to both the callers and the staff who receive calls. Each funding agency has clear eligibility rules; the HPCC systematically triages callers according to these rules so that the resulting referrals are a function of characteristics observed by the call center (and by us) and unpredictable variation in the availability of funds. As in prior studies (Evans et al., 2016; Palmer et al., 2019; Downes, Phillips, and Sullivan, 2022), we exploit these rules to compare, among callers deemed potentially eligible for funds, those referred to assistance versus those who are not. Callers who are successfully referred differ from other callers on a number of observable characteristics, but after controlling for the fund-specific rules we find that those referred to assistance have similar pre-trends in economic outcomes compared to those not referred. By linking callers to administrative tax data, we can examine whether

those who call when funding is available have different subsequent earnings and employment rates compared to those who call when funding is unavailable. Moreover, the tax data allow us to follow callers for many years both before and after they call.

We have several findings. First, our results do not support the hypothesis that families in crisis reduce their labor supply in response to temporary income transfers. For the full sample of eligible callers, our results suggest that, if anything, temporary assistance leads to greater earnings over time, although the estimated effect varies somewhat across specifications. Generally, we can reject that earnings over the following four years fall by more than 3 percent of their pre-transfer average.

Second, any long-term positive effect on earnings is driven by callers with low baseline earnings. The poorest callers are of particular interest as they may be at greatest risk of a downward spiral of unemployment and housing instability, and financial assistance has larger treatment effects for them in prior studies of emergency shelter use, violent crime arrests, and healthcare use (Evans and Moore, 2011; Palmer et al., 2019; Downes et al., 2022). For this group, we can reject the hypothesis that temporary assistance discourages work. In fact for this group, we find that being referred to financial assistance increases annual earnings over the next four years by $400-$500. This is a reasonably large effect, representing nearly 10% of the control group mean or about 55% of the total long-term effect of eviction on earnings in Collinson et al. (2024). The estimated effects on employment provide some evidence of a positive employment effect for those with low baseline earnings.

Third, looking beyond labor supply, we find that one-time emergency assistance has little impact on non-labor income sources—the effects for adjusted gross income (AGI) are consistent with those for earnings. We also do not find evidence of economically significant changes in takeup of government programs such as unemployment insurance. Fourth, we also find little evidence that the impact of financial assistance on labor market outcomes differed across groups, other than the noticeable difference by baseline earnings. Our main specification provides some evidence that earnings gains may be largest for those who are

seeking assistance because they lost their job and for those without kids. However, these heterogeneous effects are less apparent with alternative specifications.

These results have several implications. First, any work disincentives created by housing assistance appear to depend on the intensity of assistance. While some long-term subsidies have shown signs of moderate work disincentives (Jacob and Ludwig, 2012; Gubits, Shinn, Wood, Brown, Dastrup, and Bell, 2018), we can reject large work disincentives for temporary housing assistance.

Second, any improvements in labor-market outcomes resulting from stable housing appear over the long run, through the first 4 years post-treatment. This persistence, made observable by our novel data linkage, is important because even modest income gains each year, if accrued, could dramatically bolster the overall estimated net present value of emergency assistance. Also, future work on the benefits of housing support should explore effects not just at the time of award, but in the years following, as the benefits we document would have been impossible to identify with only short-term outcome data.

Together, these results support the idea that housing instability matters for labor market outcomes. If exceptionally low-income individuals lack the ability to insure against housing shocks, then these shocks can lead to negative effects in other outcomes. Our estimates suggest that, for the poorest of the poor, emergency housing assistance can help to prevent the type of downward spiral that can be associated with a negative shock.

2. Income Transfers and Labor Market Outcomes

Economists have long been interested in the labor supply effects of a wide variety of cash transfer programs including means-tested transfer programs such as Temporary Assistance to Needy Families (Ziliak, 2015), the Supplemental Nutrition Assistance Program (Han, 2022; Hoynes and Schanzenbach, 2015), and Supplemental Security Income (Duggan, Kearney, and Rennane, 2015; Deshpande, 2016). These programs can disincentivize work both because

they provide an income guarantee for those who do not work and because they apply a benefit reduction rate for those with earnings that exceed some (typically low) threshold.

Emergency financial assistance differs from these programs in important ways. Emergency financial assistance is very short-term and benefits typically do not depend on employment status or earnings level. For this reason, financial assistance can be viewed as a pure income transfer, implying no substitution effect but potentially a negative income effect. A recent literature has examined pure income transfers such as casino profits distributed to tribal members (Akee, Copeland, Keeler, Angold, and Costello, 2010), the distribution of Alaska Permanent Fund dividends to state residents (Jones and Marinescu, 2022), or expanded Child Tax Credits (Corinth, Meyer, Stadnicki, and Wu, 2021; Enriquez, Jones, and Tedeschi, 2023). The possible work disincentives have been central to debates over expanding Child Tax Credits or guaranteed income programs (Corinth et al., 2021; Hoynes and Rothstein, 2019). Our context differs from these studies because we focus on transfers that are less permanent and target a uniquely vulnerable population: those facing a housing crisis.

Labor supply effects of transfers have also been the focus of studies of housing subsidies. Jacob and Ludwig (2012) find that winning a housing voucher lottery reduces subsequent employment, although the effect is modest—a decline of 6 percent. Gubits et al. (2018) find that long-term subsidies provided to families at the moment they are entering emergency shelters reduces future employment. However, there is some evidence that the work disincentive may differ when the subsidy is temporary. The same study finds that rapid re-housing subsidies, which typically phase out within 12 to 24 months, increase earnings in the short-term (Gubits et al., 2018). But the study's power for measuring long-term effects is limited due to relatively low take-up. This is important, since one of the rationales for these temporary subsidy programs concerns their impact on future income.

Previous studies of emergency financial assistance have not focused on labor market outcomes. A series of studies examines how such assistance affects homelessness (Evans et al., 2016; Phillips and Sullivan, 2025), crime (Palmer et al., 2019), and healthcare use (Downes

et al., 2022) but does not measure employment outcomes. The one existing emergency assistance study that does measure employment shows large pre-treatment imbalances in employment that make inference difficult (Rolston, Geyer, Locke, Metraux, and Treglia, 2013). Perhaps the closest papers to the present study examine small, one-time income transfers during the COVID-19 pandemic. Jaroszewicz, Jachimowicz, Hauser, and Jamison (2022) find a small negative impact of providing $2000 on a financial index mostly composed of employment outcomes, and Pilkauskas, Jacob, Rhodes, Richard, and Shaefer (2023) find little impact of a $1000 transfer on employment. Of course, these papers study programs implemented in the unique context of a global pandemic that dramatically altered labor market outcomes, and although the transfers were targeted towards low-income households, they did not target those facing a negative income shock or other crisis.

Also, whether housing assistance encourages or discourages employment may vary across individuals. In particular, although cash transfers may lower earnings and employment through an income effect, these work disincentives might be lower for those with very little income, either because of weak labor market attachment or because the potential for improved housing stability to lead to better labor market outcomes might be greater for low-income families. In fact, previous work (Evans et al., 2016) finds that temporary financial assistance is most effective at preventing homelessness among people with the least income, and housing loss may be particularly disruptive among people facing other barriers (Collinson et al., 2024).

3. The Homelessness Prevention Call Center

This section discusses the Homelessness Prevention Call Center (HPCC); for more details see the Appendix as well as Palmer et al. (2019) and Evans et al. (2016). In 2019, hotlines that provide essential community services were available to 95% of the U.S. population (Federal Communications Commission, 2023). In Chicago, individuals and families in need

of rent or utility assistance can call the city's services hotline, 3-1-1, for help. Such calls are routed to the HPCC.

The HPCC first collects callers' personal identifying information and demographic characteristics before determining eligibility and regardless of whether any financial assistance is available. This step allows us to link eligible callers to outcome data regardless of whether the caller is referred to assistance. After collecting basic demographic information, an intake specialist collects additional information necessary to determine if the caller is eligible for assistance.

The HPCC does not disburse assistance itself. Rather, it connects callers to government and private entities, or delegate agencies, that administer assistance. If a caller is determined to be eligible, the HPCC then checks to see if funding is available at one of the delegate agencies. The staff member works through a pre-set ordered list to see if any of the agencies has funds available immediately and if the caller meets any fund-specific eligibility criteria including request type (rent assistance, security deposit, or utilities), amount of assistance needed, veteran status, receipt of housing subsidies, and whether the total debt exceeds one month of rent. For example, some funds have more restrictive payment limits or may pay only for back rent, not security deposits or utilities. If an agency with funds is identified, the HPCC refers the caller to that delegate agency. The delegate agency is then tasked with meeting the household's net financial need, most frequently one month of back rent. The assistance is temporary, meeting only the immediate need and typically disallowing returns for more assistance in the following year. If an agency is not identified for financial assistance, the caller is referred to non-financial assistance.

On any given day, the availability of funds is sporadic and hard to predict. New delegate agencies come online and existing agencies shut down routinely. Also, funding may not be available continuously at currently operating agencies because they may temporarily run out of funds or are unable to distribute them immediately. Whether funds can be distributed on any given day, or time within a day, depends on many factors. For example, some delegate

agencies require that callers meet with a financial counselor before receiving assistance, and a HPCC intake specialist will not refer a caller for assistance if an interview slot is not available at the time of the call. For some agencies, there are only a fixed number of appointments available each week or month, but the availability of slots may vary due to cancellations or changes in staff capacity. Variation in funding also results from the fact that some delegate agencies are supported by local or state programs that provide an inconsistent and unpredictable funding stream. Therefore, an eligible caller who calls when delegate agencies happen to have funding and staff availability will be referred to assistance, while the same caller would have been turned away had they called when funding was unavailable or staff lacked capacity to process a request. Figure 1 shows that the weekly funding rate varies considerably, with nearly all eligible callers being referred in some weeks, but only about a quarter in other weeks.

Because fund availability is determined by factors outside the control of the HPCC, the call center has limited information about when funds will be available. Any information they might have about future funding is not shared with callers. The instructions for HPCC staff state, "If anyone asks, 'when will a fund be available?' please respond the following: 'I do not have information on when funds will be available. Unfortunately, there are not enough funds for everyone who needs assistance and availability is sporadic.' If anyone asks, 'should I call back?' please reply: 'That is up to you.' If anyone asks, 'but what is the best time to call?' please reply: 'There is no best time to call. The need is so high in Chicago/the Suburbs, there are so many people trying to get access to the limited number of grants'" (HPCC, 2013). As a result, the HPCC process generates a natural comparison group. Observationally identical callers–i.e. those with the same fund-specific eligibility criteria–are sometimes referred to funds and other times not.

4. Data and Methodology

4.1. Data

Following Downes et al. (2022), this study uses the sample of all calls received by the HPCC between July 1, 2013 and December 31, 2015. Prior to matching this sample to outcome data, we exclude calls in categories unrelated to financial assistance, e.g. calls that are incomplete or do not include client interaction. The call data include personal identifiers (e.g., name) fundamental to the data linking process, as well as characteristics related to the program eligibility process. These variables include those related to overall eligibility (e.g. income), fund-specific eligibility (e.g. amount needed), and general demographic information (e.g. gender). We construct outcome measures using data from three types of tax forms: form W-2 (used to measure wages), form 1040 (used to measure Adjusted Gross Income, or AGI), and various information returns (e.g., 1099 forms). Appendix A2 discusses our use of these forms and linkage to the HPCC data in more detail.

For our analysis, we restrict the sample of callers to those who: are eligible for assistance, can be matched to the tax data, and have no prior call in the past 6 months. In Table 1 we report mean characteristics for various samples. Comparing all HPCC callers (column 2) to people living in the same Census block as any HPCC caller (column 1) shows, not surprisingly, these are very different samples–callers are younger and much more likely to be female and less likely to be white.[1] The results in Table 1 also show that the various subsets of callers considered have similar characteristics. To test whether the lowest-income households benefit most from the intervention, in the analyses below we estimate effects separately for low- and high-earnings households using pre-period earnings. Specifically, we split the sample into low-earnings and high-earnings groups based on whether individuals' average earnings over the five years prior to the call are below or above the sample median of about $6,000. The low-wage group characteristics are in column 5.

[1] In Appendix A3 we discuss differences between columns in more detail.

4.2. Regression Specifications

We will take several approaches to estimate the effect of funding. First, we will regress outcomes on referral to funds and our controls one year at a time:

$$Y_{i\tau} = \alpha_\tau + Funds_{i0}\beta_\tau + X_{i0}\Gamma_\tau + Z_{it}\Pi_\tau + Y_{i,-1}\delta_\tau + \epsilon_{i\tau} \tag{1}$$

where $Y_{i\tau}$ is an outcome such as subsequent earnings for individual i, $Funds_{i0}$ is an indicator for whether the person was referred to funds at the time of the call (denoted year zero), calendar time is indexed by t, event year is indexed by τ, and $Y_{i,-1}$ denotes the outcome observed for individual i in the year prior to calling.[2] Z_{it} is the set of caller characteristics at the time of the call, such as need amount, that affect fund-specific eligibility and the probability of referral, as discussed in Section 3.[3] It is indexed by calendar time t to capture the fact that the maximum amount offered by different delegate agencies changes over the sample period; we allow for interactions of need amount categories with calendar year-quarter. We further include X_{i0}, a set of controls for caller characteristics observed at baseline but not used by delegates, to reduce residual variance in the outcome.[4] A year-by-year estimation of (1) allows β_τ (and all other coefficients) to change each year in an event-study-style analysis.

Second, we will pool all post-referral observations together and estimate β using all post-referral years at once:

$$Y_{i\tau} = \alpha + Funds_{i0}\beta + X_{i0}\Gamma + Z_{it}\Pi + Y_{i,-1}\delta + \epsilon_{i\tau} \tag{2}$$

estimated over $\tau \geq 0$, a "pooled-event-study analysis".

Third, since our administrative tax data allows us to track individuals over time, we

[2]When a caller has made multiple calls that satisfy our sample conditions within our analysis period, each call record is included in our analysis. For the purposes of event study analysis, each call has relative year set to zero in the calendar year it was made.

[3]See notes of Table 3 for a full list of variables in Z_{it}.

[4]See notes of Table 3 for a full list of variables in X_{i0}.

consider individual-fixed-effect specifications. These regressions use all years of data both pre- and post-referral and interact the referred-to-funds dummy with an indicator for whether a given year is post-referral:

$$Y_{i\tau} = \alpha + Funds_{i0} \times 1_{\tau \geq 0}\ \beta + Z_{it}\Pi + \mu_i + \iota_\tau + \kappa_t + \nu_a + \epsilon_{i\tau} \tag{3}$$

where $1_{\tau \geq 0}$ is a post-referral indicator, the individual-specific effects μ_i will preclude estimation of coefficients for the $Y_{i,-1}$ control (along with a not-interacted $Funds_{i0}$ coefficient and the X and Z controls that are time-invariant), event-year effects ι_τ subsume a not-interacted $1_{\tau \geq 0}$ variable, κ_t is a fixed effect for calendar year, and ν_a is a fixed effect for age in years.

These equations, while conventional, use linear covariate controls. Recent work has noted that this assumption may cause a bias in the estimate of β if referral decisions are in fact nonlinearly related to covariates (see section 3.8 in Mogstad and Torgovitsky (2024); see also Abadie (2003) and Słoczyński, Uysal, and Wooldridge (2025)). Our final approach will address this issue by using the pooled-event-study approach in (2) while introducing a much more comprehensive set of interactions of the controls for the observable Z_{it} terms.

Ideally, we would estimate a fully saturated model, including fixed effects for all unique values from interactions of the Z variables with each other and a dummy representing a window of time (e.g., a given quarter in a given year). Intuitively, this would identify the effect of referral by comparing observationally equivalent individuals, exploiting our exogenous variation as much as possible. This would also avoid biases from incorrectly imposing linearity in our controls.[5] Below we present estimates from a fully-saturated model.[6] However, we have several thousand unique values of the vector Z interacted with time windows, and estimates from a fully saturated model are imprecise. So, we also approximate this fully

[5]This "rich covariates" approach also has the consequence that an IV estimand can be constructed as a weighted average of $\beta(z)$ terms, where each $\beta(z)$ is simply the linear IV estimand with no covariates conditional on $Z_{it} = z$; see equation (29) in Mogstad and Torgovitsky (2024). We do not observe whether a caller eventually receives funding, so our results are ITT, as discussed below.

[6]One of the Z variables, rank of call within the day, is unique to individuals and cannot be fully discretized; we discretize it into quartiles.

saturated specification by estimating a more parsimonious specification, interacting our calendar quarter-by-year dummies with dummy variable values of our Z_{it} controls (but not the Zs with each other). In addition, we estimate a specification that relies on a double selection LASSO (Belloni, Chernozhukov, and Hansen, 2014) approach that selects controls from the fully saturated set of interactions. For the low-wage group, our main coefficients are similar across all of these approaches to saturation.

The key assumption behind all of these approaches is that referral to funding is conditionally exogenous. We assess the validity of this assumption by comparing baseline characteristics of those referred and not referred to funds, controlling for characteristics related to fund-specific eligibility. In particular, we estimate for each year-zero characteristic X_{i0} at time $\tau = 0$:

$$X_{i0} = \phi + Funds_{i0}\psi + Z_{it}\Omega + u_{i0} \tag{4}$$

The top panel of Table 2 shows results with each row corresponding to a different X_{i0} variable. Group means and raw differences are in the first three columns. The 'adjusted difference' column reports the estimate of ψ. Some estimates are statistically different from zero, and an F-test rejects the joint null of balance across all X_{i0} (p=0.00). These differences are clearly non-random and likely due to departures of our main linear specification from a fully saturated model. However, they are likely inconsequential. First, these adjusted differences are small. For example, those referred to funds have \$260 greater monthly income but conditional on Z_{it} this difference becomes only $-$\$57 per month. Second, they are not systematic. Conditional on Z_{it}, referred individuals report slightly lower monthly income to the HPCC, but, as shown in the second panel, have greater lagged wages reported on W-2 forms.

The final panel of the table provides summary measures of balance: predicted earnings and income. Using the non-referred group only, we regress year-two earnings and AGI on all the baseline caller characteristics in Appendix Table A.7, including those collected by the HPCC and measures of lagged wages and AGI. We then use these regressions to

predict earnings and AGI for the treatment and control groups. As the last column shows, the differences in predicted earnings or AGI are small and not statistically significant after characteristics related to fund-specific eligibility are accounted for. Overall, we find that referral occurs for similar looking households, with similar predicted future values (absent treatment) in outcomes and similar pre-trends (as discussed below). This baseline balance is particularly strong for our low-earnings subsample (see appendix Table A.8). For a more detailed discussion of the evidence on baseline balance see Appendix A.3; baseline values of other variables (including some Z_{it} variables) are shown in Appendix Table A.7.[7]

Our estimates are intent-to-treat (ITT); we do not observe whether a caller eventually receives funding. Prior studies indicate that 71 percent of callers approved for funding received/expected to receive assistance or had a request in process (George, Hilvers, Patel, and Guelespe (2011)). In our data 5.4% of those not referred call back later and are referred, suggesting 4% (71% $\times$ 5.4) of our control group would receive assistance through repeat calling. George et al. (2011) find that 8.4% of callers turned away by HPCC had their need met elsewhere. Taken together, this suggests roughly 12% of those not referred eventually receive assistance. The difference between assistance among the treated (71%) and among the control group (12%) suggests that those initially referred are 59 percentage points more likely to receive assistance. A 2SLS estimate of treatment effects would thus be about 69% (or 1/0.59) larger than the ITT effects we report.

5. Results

Figure 2 shows trends in earnings in the five years prior to an individual calling the HPCC (denoted year 0) and in the four years after. We report these trends for the full sample, as well

[7]To provide additional evidence that referral to funds is conditionally exogenous, we can also exploit information on the reason an individual calls the HPCC, since different funds address different needs. We calculated the fraction of eligible individuals referred to funds in a given year/quarter/Z_{it} cell, and regressed this fraction on the fraction of all *other* individuals that quarter referred to funds. The coefficient is small and insignificant (0.0823, se = 0.0821), indicating our controls identify groups of individuals wherein referral rates are unrelated to the referral rates of others.

as separately for our low- and high-earnings subsamples. Panels a-c report raw, unadjusted mean earnings, while panels d-f report regression adjusted means that account for observable characteristics (both X_{i0} and Z_{it}). The trends in panels a-f show rising earnings over time for both those referred and not referred to funds, but differences in the levels and the trends in pre-call earnings, especially for the high-earnings group. Those referred to funds have higher annual earnings than those not referred and earnings that rise faster. The flattening of earnings for the treatment group in period zero in the upper left figure is driven entirely by the high-earnings sample, where there is a large drop in earnings in year zero. This drop resembles the "Ashenfelter dip" in earnings observed for evicted individuals in Collinson et al. (2024), though they compare evicted to non-evicted individuals, while we compare individuals who (voluntarily) call the HPCC and are referred to funds to callers who are not referred. Importantly, both this pre-trend and the period-0 decline are absent for the low earnings group.

The rightmost three panels present event-study-style estimates based on year-by-year regressions of equation (1). Each coefficient is the estimate β_τ taken from a regression of wages on a dummy for whether an individual is ever referred to funds, along with the controls X_{i0}, Z_{it}, and lagged wages. Robust standard errors are used to construct the confidence intervals.

Several things are noteworthy. First, the results are quite different after controls are introduced. The increasing pre-trend for the high earnings group in the bottom row is eliminated, and the pre-treatment gap in earnings between those referred and not-referred to funds is much smaller and insignificant. Next, there are no significant differences in post-treatment earnings for the high-earnings group, but there are for the low-earnings group, where earnings appear to be higher following referral even five years later. The large gains in earnings two years after referral indicate a *positive* effect on earnings for this group.

Table 3 consolidates these treatment results using several different specifications. Standard errors are clustered by caller, and each reported number of observations is rounded

to preserve anonymity. The 'No Lag' column reports estimates from equation (2) excluding the lagged outcome. The resulting coefficient is a pooled-event-study estimate.[8] The 'Main' column estimates equation (2), including the pre-calling control $Y_{i,-1}$, the 'No Xs' column includes $Y_{i,-1}$ but excludes X_{i0}, and the 'Fixed Effects' column uses individual fixed effects as in equation (3). The last three columns estimate equation (2) but include our much stronger set of interaction controls. 'Zs x Qtr' interacts each Z_{it} control with quarter-by-year dummies. The low-wage-sample estimates are very close to before, while the estimates for all individuals are closer to zero and not statistically significant. The next column is fully saturated, adding fixed effects for unique interactions of all Z_{it} variables with each other, year, and quarter. The estimates are notably close to those in the main specification, while the standard errors are unsurprisingly larger. The final column selects controls from the fully saturated model using a LASSO, and the coefficients are for all samples positive and significant.

These various specifications consistently reject large decreases in earnings. In the overall sample, all the estimates are positive, suggesting that being referred to assistance increases earnings, but the magnitude and precision is sensitive to the specification. The full-sample significant positive effect when we omit a control for lagged earnings is consistent with some callers with higher earnings in period -1 being more likely to see referral (as suggested by Figure 2.c). When we include lagged earnings (column 2) the estimated effect is much smaller (197) and not significant. But we can reject that earnings decrease by more than \$353, or 3% of average pre-call earnings. The fixed effect specification point estimates are much larger and can reject any decline as can the LASSO-based specification.

Any positive earnings effects we observe from referral are strongest for low-earnings callers. For this group, the effects are positive for all specifications and suggest long-term increased earnings of \$400-\$500 from referral, consistent with Figure 2.[9] This represents nearly

[8] This specification follows the approach used in earlier papers that study the effects of financial assistance using the same context as ours (Evans et al., 2016; Phillips and Sullivan, 2025; Palmer et al., 2019).

[9] In an earlier draft we also reported results that used inverse-propensity weights based on the likelihood of referral; the low-wage sample estimates were similar with weights, both with fixed effects and without.

10% of the control group mean for low earners and about 55% of the total long-term effect of eviction on earnings in Collinson et al. (2024).[10] The estimates are at or near statistical significance (the p-value in the second column is about 0.104).[11] For all specifications we can reject the hypothesis that being referred to funds leads to a decline in earnings, which is an important result.[12] The bottom panel of Table 3 reports our main estimates for the high earnings sample. For this group, the effect of being referred to assistance is more sensitive to the specification used, and the effect is only significant in the LASSO-based specification.

The appendix presents several extensions to these results. First, Table A.1 shows results from our models interacting Zs and time controls with and without the lagged outcome control $Y_{i,-1}$; removing this generally produces similar results, especially for the low-wage estimates. Next, Table A.2 replaces quarter-based time dummies with month- and week-based dummies for the 'Zs x Qtr' specification of Table 3. The results, especially for the low-wage group, are robust to this variation.

Next, Table A.3 considers employment, total tax-unit wages, adjusted-gross income, and indicators for whether an individual files a 1040 or receives a 1099-G or 1099-MISC. The coefficients are consistently positive for the low-wage group, suggesting better labor market outcomes across these measures, although the results are only consistently statistically significant under the LASSO specification. But across specifications the low-wage group

We omit those results here for brevity but note that low-wage-sample estimates including dummies for interactions of the Zs with time-period dummies generally produce similar results when weights are used.

[10] Collinson et al. (2024) report quarterly effects of \$323 in the first year, \$613 in the second (Table 6), and \$217 in the third & fourth (Appendix table H6), for an average of \$342.5. Adjusting a \$450 ITT effect by a factor of 1.69 to make it TOT and dividing by 4 to make it quarterly produces \$191.1; $191/343 \approx .55$.

[11] For the specifications with well-defined strata, we can also conduct randomization-inference, where we reassigned treatment within $Z_{it} \times$Qrtr cells 1,000 times and generated a p value by comparing the true estimate to the placebo-estimate distribution. For the fully saturated model, estimates of the main- and low-wage-sample p values were 0.22 and 0.54, respectively, compared to 0.18 and 0.49 with analytical standard errors. For LASSO, the randomization-inference p-values are 0.041 and 0.021, respectively.

[12] Figure A.1 explores heterogeneity by earning status further. The figure depicts two plots of 101 regressions where the dependent variable is an indicator for an individual reporting earnings greater than x, for x values ranging from \$0 to \$100,000. Panel (a) uses a pooled-event-study estimation of equation (2); panel (b) uses an individual fixed-effects specification as in column 4 of Table 3 and equation (3). The plot in panel (b) is negative to the left of the figure while the panel (a) results are smaller, positive at zero, and highly insignificant. Both plots become positive around \$15,000 and in both cases estimates at very high levels (near \$80,000) are marginally significant, suggesting a small (about 0.2 percent) probability of large earnings gains (or, equivalently, the avoidance of large earnings losses) from emergency assistance.

consistently has (a) higher AGI and (b) a greater likelihood of filing a 1040. The results rule out large decreases in employment and also suggest that earnings increases for low-wage earners do not crowd out other household earnings or sources of AGI.

Last, Tables A.4 and A.5 split the sample by (a) reason for calling the HPCC—job loss versus any other reason—and then (b) by having kids or not having kids, as reported to the call center at the time of the call.[13] Splitting the sample unsurprisingly lowers the precision of the estimates. The earnings and AGI results suggest higher benefits to those seeking help because they have lost a job, and the results also suggest larger benefits for those without kids, but this is not seen in all specifications. This raises the possibility that greater fund availability to those seeking help from job loss, and those without kids, could have relatively large economic benefits. This result parallels prior work by Palmer et al. (2019) showing that the same intervention's violent crime reduction benefits operate primarily through households without children.

6. Conclusion

This study uses federal tax records to measure earnings for people who request assistance from a call center after they have experienced a negative shock that puts their housing at risk. Due to external constraints on funding and staffing, the call center refers some callers to temporary financial assistance but not others, generating quasi-random variation in assistance. Exploiting this variation, we find that this targeted and temporary assistance does not reduce earnings. For the full sample, the point estimate is positive, and we can reject that earnings fall by more than 3% of the baseline mean. Moreover, we find evidence that this financial assistance increases earnings by \$400-\$500 among people with earnings below the sample median at baseline.

Our results imply that existing analyses of homelessness prevention programs underesti-

[13] We have considered other dimensions of heterogeneity including marital status, gender, age, receipt of benefits, and whether it is a first-time call, among others. These results do not generate clear differences in the effect of assistance across groups.

mate their benefits. Existing studies (Rolston et al., 2013; Evans et al., 2016; Phillips and Sullivan, 2025) demonstrate that such programs stabilize housing. While studies of other, more permanent, housing subsidies might lead to concerns of disemployment effects (Jacob and Ludwig, 2012), our results suggest any disemployment effect is small for temporary assistance. In fact, our results suggest that temporary assistance encourages employment for the lowest earners. For this group, the present value of increased earnings far exceeds the cost of the program. As a result, existing cost-benefit analyses (Phillips and Sullivan, 2025) that value the private benefits of financial assistance at the payment amount are conservative.

More generally, our results suggest that many low-income households are underinsured against shocks to their income. If future labor market earnings can pay for the cost of temporary financial assistance, then households likely face barriers to credit and insurance that would otherwise allow them to move resources across time or states of the world. Some have argued that housing is a necessary platform for stable employment, and temporary disruptions to housing can permanently shift people into poverty (Desmond, 2016). Our results suggest that, to the extent that this is true, providing insurance against temporary shocks can help to undo such a poverty trap.

References

Abadie, A., 2003. Semiparametric instrumental variable estimation of treatment response models. Journal of Econometrics pp. 231–263.

Akee, R. K., Copeland, W. E., Keeler, G., Angold, A., Costello, E. J., 2010. Parents' incomes and children's outcomes: a quasi-experiment using transfer payments from casino profits. American Economic Journal: Applied Economics 2, 86–115.

American Rescue Plan Act, 2021. U.S. House of Representatives. 117th Congress, 1st Session.

Belloni, A., Chernozhukov, V., Hansen, C., 2014. High-dimensional methods and inference on structural and treatment effects. Journal of Economic Perspectives 28, 29–50.

CARES Act, 2020. 15 U.S.C. § 9001.

Cesarini, D., Lindqvist, E., Notowidigdo, M. J., Östling, R., 2017. The effect of wealth on individual and household labor supply: evidence from Swedish lotteries. American Economic Review 107, 3917–3946.

Collinson, R., Humphries, J. E., Mader, N., Reed, D., Tannenbaum, D., Van Dijk, W., 2024. Eviction and poverty in american cities. The Quarterly Journal of Economics 139, 57–120.

Consolidated Appropriations Act, 2020. U.S. House of Representatives. 116th Congress, 2nd Session.

Corinth, K., Meyer, B. D., Stadnicki, M., Wu, D., 2021. The anti-poverty, targeting, and labor supply effects of replacing a child tax credit with a child allowance. Tech. rep., National Bureau of Economic Research.

Deshpande, M., 2016. The effect of disability payments on household earnings and income: Evidence from the SSI children's program. Review of Economics and Statistics 98, 638–654.

Desmond, M., 2016. Evicted: Poverty and profit in the American city. Crown.

Downes, H., Phillips, D. C., Sullivan, J. X., 2022. The effect of emergency financial assistance on healthcare use. Journal of Public Economics 208, 104626.

Duggan, M., Kearney, M. S., Rennane, S., 2015. The Supplemental Security Income program. In: *Economics of Means-Tested Transfer Programs in the United States, Volume 2*, University of Chicago Press, pp. 1–58.

Enriquez, B., Jones, D., Tedeschi, E., 2023. Short-term labor supply response to the expanded Child Tax Credit. In: *AEA Papers and Proceedings*, American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, vol. 113, pp. 401–405.

Evans, W. N., Moore, T. J., 2011. The short-term mortality consequences of income receipt. Journal of Public Economics 95, 1410–1424.

Evans, W. N., Sullivan, J. X., Wallskog, M., 2016. The impact of homelessness prevention programs on homelessness. Science 353, 694–699.

Federal Communications Commission, 2023. Dial 211 for essential community services.

George, C., Hilvers, J., Patel, K., Guelespe, D., 2011. Evaluation of the Homelessness Prevention Call Center. Loyola U Chicago Center for Urban Research and Learning (CURL) Report .

Golosov, M., Graber, M., Mogstad, M., Novgorodsky, D., 2024. How americans respond to idiosyncratic and exogenous changes in household wealth and unearned income. The Quarterly Journal of Economics 139, 1321–1395.

Gruber, J., 1997. The consumption smoothing benefits of unemployment insurance. American Economic Review 87, 192.

Gruber, J., 2000. Cash welfare as a consumption smoothing mechanism for divorced mothers. Journal of Public Economics 75, 157–182.

Gubits, D., Shinn, M., Wood, M., Brown, S. R., Dastrup, S. R., Bell, S. H., 2018. What interventions work best for families who experience homelessness? Impact estimates from the family options study. Journal of Policy Analysis and Management 37, 835–866.

Han, J., 2022. The impact of SNAP work requirements on labor supply. Labour Economics 74, 102089.

Hoynes, H., Rothstein, J., 2019. Universal basic income in the United States and advanced countries. Annual Review of Economics 11, 929–958.

Hoynes, H., Schanzenbach, D. W., 2015. US food and nutrition programs. In: *Economics of means-tested transfer programs in the United States, volume 1*, University of Chicago Press, pp. 219–301.

HPCC, 2013. Homelessness Prevention Call Center Script Guidelines. Homelessness Prevention Call Center Document .

Hungerman, D., Phillips, D., Rinz, K., Sullivan, J., Wasser, D., 2025. The effect of emergency financial assistance on mobility, snap receipt, and presence of dependents. American Economic Review Papers / Proceedings .

Hungerman, D. M., Phillips, D. C., Rinz, K., Sullivan, J. X., 2024. The effect of emergency financial assistance on employment and earnings. NBER Working Paper 32856 .

Imbens, G. W., Rubin, D. B., Sacerdote, B. I., 2001. Estimating the effect of unearned income on labor earnings, savings, and consumption: Evidence from a survey of lottery players. American Economic Review 91, 778–794.

Jacob, B. A., Ludwig, J., 2012. The effects of housing assistance on labor supply: Evidence from a voucher lottery. American Economic Review 102, 272–304.

Jaroszewicz, A., Jachimowicz, J., Hauser, O., Jamison, J., 2022. How effective is (more) money? Randomizing unconditional cash transfer amounts in the US. Unpublished working paper .

Jones, D., Marinescu, I., 2022. The labor market impacts of universal and permanent cash transfers: Evidence from the Alaska Permanent Fund. American Economic Journal: Economic Policy 14, 315–340.

Layne, M., Wagner, D., Rothhaas, C., 2014. Estimating Record Linkage False Match Rate for the Person Identification Validation System, cARRA Working Paper No. 2014-02.

Mogstad, M., Torgovitsky, A., 2024. Instrumental variables with unobserved heterogeneity in treatment effects. NBER Working Paper .

Mulrow, E., Mushtaq, A., Pramanik, S., Fontes, A., 2011. Assessment of the U.S. Census Bureau's Person Identification Validation System. Tech. rep., NORC at the University of Chicago.

Olivet, J., 2023. Keynote Remarks to the HUD Office of Policy Development and Research Quarterly Updated on Housing First.

Palmer, C., Phillips, D. C., Sullivan, J. X., 2019. Does emergency financial assistance reduce crime? Journal of Public Economics 169, 34–51.

Phillips, D. C., Sullivan, J. X., 2025. Do homelessness prevention programs prevent homelessness? Evidence from a randomized controlled trial. The Review of Economics and Statistics 107, 1187–1196.

Pilkauskas, N. V., Jacob, B. A., Rhodes, E., Richard, K., Shaefer, H. L., 2023. The COVID cash transfer study: The impacts of a one-time unconditional cash transfer on the well-being of families receiving SNAP in twelve states. Journal of Policy Analysis and Management 42, 771–795.

Price, D. J., Song, J., forthcoming. The long-term effects of cash assistance. American Economic Journal: Applied Economics .

Robins, P. K., 1985. A comparison of the labor supply findings from the four negative income tax experiments. Journal of Human Resources pp. 567–582.

Rolston, H., Geyer, J., Locke, G., Metraux, S., Treglia, D., 2013. Evaluation of the Homebase Community Prevention Program. Final Report, Abt Associates Inc, June 6, 2013.

Słoczyński, T., Uysal, S. D., Wooldridge, J. M., 2025. Abadie's kapp and weighting estimators of the local average treatment effects. Journal of Business & Economic Statistics pp. 164–177.

Wagner, D., Layne, M., 2014. The Person Identification Validation System (PVS): Applying the Center for Administrative Records Research and Applications' (CARRA) Record Linkage Software, cARRA Working Paper No. 2014-01.

Ziliak, J. P., 2015. Temporary Assistance for Needy Families. In: *Economics of Means-Tested Transfer Programs in the United States, Volume 1*, University of Chicago Press, pp. 303–393.

Appendix

A1. The HPCC Center: further details

To be eligible, a caller must demonstrate the presence of a crisis that (a) is on a list of eligible crises (such as the loss of a job), (b) causes imminent risk of homelessness (as evidenced by an eviction notice, for example), (c) can be solved with limited financial assistance (typically less than $1,500), and (d) is temporary (i.e. the caller has sufficient future income to pay essential expenses). The center initially screens on these criteria based on information the caller provides, but funding agencies verify many of these eligibility criteria (e.g. with landlord and employer documentation) before funding is provided.

Callers ineligible for the HPCC and callers who are not referred to funds are referred to non-financial assistance. This assistance includes legal aid, domestic violence counseling, assistance with utility complaints, workforce development, senior services, disability services, public benefit screening, and other general support services (George et al., 2011).

Figure A.2 shows that the probability of funding varies considerably from week to week. While the average funding rate for this sample period was 65%, in some weeks more than 90% of eligible callers were referred to funds, while in other weeks less than 30% were referred. This variation may differ across different caller types because of fund-specific eligibility requirements. Results look similar when we limit the sample to a shorter time period during which we observe fund-specific eligibility requirements (Figure A.2.b). In this shorter time period, we can restrict the sample to callers that share some of the same fund-specific eligibility requirements, i.e. callers who do not have other housing subsidies and are asking for more than one month of rent assistance totaling more than $900 (A.2.c), we still find considerable variation in the likelihood of being referred to funds.

A2. Administrative Data Details

We construct outcome measures using data from three types of tax forms: form 1040, form W-2, and various information returns (e.g., 1099 forms). In each case, we have access to a limited set of fields from the universe of records. Form 1040 is available annually starting in 1998 and provides tax unit-level information on income, filing status, presence of children, and place of residence. Because we have access to the universe of returns, the data also tell us whether a person appears on any tax return in a given year. Form W-2 is available annually beginning in 2005 and provides individual-by-employer-level information on earnings. Our primary analysis aggregates earnings across employers within person. In addition to analyzing earnings, we use data from W-2s to construct measures of employment, with

people considered employed if they have earnings greater than zero, and more substantially employed if they have earnings exceeding the amount associated with half a year of full-time work at the contemporaneous federal minimum wage.

The information return data includes whether a person receives various information returns and the addresses at which forms were received. The address information is available beginning in 2003. Codes identifying the type of information return received are available consistently beginning in 2010.[14] The US Census Bureau's Numident file provides demographic information for every person with a social security number. Information provided includes date of birth, place of birth, sex, and citizenship.

For our comparisons of callers to non-callers, we also use data from the decennial census and American Community Survey (ACS) to provide other demographic data at the individual level, most notably information on race and ethnicity. This information is drawn from the 2000 (short form) and 2010 decennial censuses and all available years of the ACS, which samples one percent of households annually beginning in 2005. We determine place of residence for callers and non-callers using location information available from tax forms and the Census Bureau's Master Address File Auxiliary Reference File (MAF-ARF) which collects person-year-address combinations from additional administrative data sources.

The HPCC data are linked to outcome data at the individual level using the Census Bureau's Protected Identification Keys (PIKs), assigned using personally identifiable information (PII) by the Person Identification Validation System (PVS).[15] Each PIK is associated with a unique social security number (SSN) or individual taxpayer identification number (ITIN). For this reason, files that include information on people's SSNs/ITINs (e.g. administrative records such as data from tax forms) typically have extremely high PIK assignment rates (97 percent or higher). PIK assignment rates for the decennial census and ACS, which do not contain SSN/ITIN information, are typically somewhat lower but still high in absolute terms at about 90 to 93 percent (Mulrow, Mushtaq, Pramanik, and Fontes, 2011). PVS's false match rates are generally extremely low (Layne, Wagner, and Rothhaas, 2014).

In the HPCC data, about 95 percent of records are successfully assigned a PIK, in line with the high PIK assignment rates for the other data we are using. For callers who meet the criteria to be included in our main analysis sample, Table A.6 compares those who are assigned a PIK to those who are not, using information collected by the HPCC. In general, the differences between the two groups are fairly small. One notable exception is that the people not assigned a PIK are nearly three times as likely to be Hispanic as people who are

[14] The specific information returns identified by these codes are 1099-MISC, 1099-R, 1099-S, 1099-G, 1099-SSA, 1099-INT, 1099-DIV, and 1098. The file also includes an indicator for receipt of and address information associated with Form W-2.

[15] See Wagner and Layne (2014) for details.

assigned a PIK (21 percent vs. 7.5 percent). However, given the small size of the group not assigned PIKs, their exclusion from the sample does little to shift the baseline characteristics of the HPCC callers whose subsequent outcomes we are able to analyze.

A3. Referred callers, Not Referred Callers, and Non-callers

Because we have access to the universe of records from the data sources described in Section 4.1, we can compare HPCC callers to non-callers from the same narrow geographic area based on contemporaneous and pre-call characteristics. This comparison, which has not been feasible in prior studies of this program, improves our understanding of who seeks emergency financial assistance, and specifically who calls the HPCC.

We start broadly, comparing all callers, regardless of whether they are eligible for assistance, to non-callers. The first two columns of Table 1 compare HPCC callers from 2013 through 2015 to non-callers who were living in the same set of Census blocks at the time.[16] The comparison must be based on information that is available for both groups and so is limited to demographic characteristics obtained from full population data available through the Census Bureau. There are large differences on several characteristics between callers and their geographic neighbors. Callers are three years younger on average, 25 percentage points more likely to be women, and nearly 21 percentage points more likely to be Black than are non-callers living in the same Census blocks. Callers are also nearly 14 percentage points less likely to have been born outside the United States, and correspondingly more than 16 percentage points more likely to be citizens.

Columns 3 through 5 show how we construct our analysis sample from the call data and how caller characteristics change as we narrow the sample. These restrictions are identical to those in Downes et al. (2022). We first limit the sample in column 3 to people who meet general eligibility requirements: experienced a crisis, at imminent risk of homelessness, solvable with limited financial assistance, and demonstration of future income (see program description above). Because people can call repeatedly and likely do so non-randomly, we limit our sample to calls with no prior call in the past 6 months. Column 4 shows the set of people with a call in our main analysis sample. Finally, at times we analyze data separately for callers with the lowest pre-period wages; column 5 shows the sample of callers with pre-period wages below the median. While there are some small fluctuations on these characteristics across subsets of callers, these changes are small compared to the gaps between

[16] Census blocks are the smallest geographic unit used by the Census Bureau for official data tabulations. Their boundaries are generally formed by streets and highways, especially in more urban areas where they often consist of city blocks, but may also consist of landscape features such as bodies of water or political borders, such as county/city/town boundaries.

the full set of callers and non-callers living in the same places. Even when we restrict the sample to those with low earnings, which is a sample we focus on in our analysis, the observable characteristics are quite similar to those for the full sample of eligible callers.

We also consider figures that match eligible callers to non-callers using propensity score matching. Propensity scores are estimated using a logistic regression of calling HPCC on the characteristics shown in the table. Matches are drawn without replacement. Where possible, each caller's match is selected from the same Census block as the caller.[17] We use similarly identified matched samples to examine differences between callers and observably similar non-callers on time-varying, pre-call characteristics. In Table 1 of (Hungerman, Phillips, Rinz, and Sullivan, 2024) we show that differences between the HPCC low-wage callers and the matched sample on the characteristics used in the matching process are fairly small.

Figure A.3 shows trends in select outcomes for all HPCC callers in 2013–2015 and matched non-callers, beginning at least ten years before a call.[18] There are clear differences between these two groups: HPCC callers are less likely to be employed (i.e. to have any income reported on a W-2), have lower AGI, claim more children as dependents on their tax returns, and change addresses more frequently than demographically similar non-callers living in the same parts of Cook County. These trends do not point to particularly sharp changes in circumstances preceding callers' interactions with the HPCC, though the number of children claimed on tax returns does trend up for callers over the preceding years, with a notable jump up five years prior, while trending down for matched non-callers, suggesting that the presence of children could contribute to financial need for people seeking assistance from HPCC.

Trends are similar for our full analysis sample and the corresponding matched non-callers (Figure A.4), though the difference in the likelihood of being employed is notably smaller for this sample. The smaller gap between callers in our analysis sample and matched non-callers with respect to employment likely reflects the screening done by the HPCC to ensure that aid is provided to people facing a temporary crisis that causes an imminent risk of homelessness but can be solved with limited financial assistance. The fact that potential contributors to financial need such as the presence of children do not meaningfully change relative to non-callers between Figure A.3 and A.4 while employment gaps close suggests that callers with relatively weak employment prospects are screened out by the HPCC, likely because a modest amount of financial assistance is less likely to fully meet their financial obligations.[19]

[17] If matching within block is not possible, the match is selected from the same tract. If matching within tract is not possible, the match is selected randomly from non-callers living in the union of all blocks that were home to HPCC callers that year.

[18] For non-callers, the year of the call is the year in which they are matched to a caller.

[19] Hungerman, Phillips, Rinz, Sullivan, and Wasser (2025) has a brief discussion of callers and non-callers

Table A.7 compares those referred and not referred to funds. The sample is restricted to eligible callers, with a PIK, who have not called in the prior six months and with nonmissing characteristic information. The fourth column in Table A.7 shows differences in characteristics after controlling for characteristics related to fund-specific eligibility and the timing of the call, Z_{it}. In the last two rows of the table, we compare predicted earnings and income for those referred and not referred. Using the non-referred group only, we regress year two earnings and AGI on all the baseline caller characteristics in the table collected by HPCC as well as measures of lagged wages and AGI. We then use these regressions to predict earnings and AGI for the treatment and control groups. Comparisons of these predicted outcomes provide a summary measure of balance, as they capture a weighted average of how differences in observable characteristics across groups might affect our key outcomes. As the last column shows, the differences in predicted earnings or AGI are small and not statistically significant after characteristics related to fund-specific eligibility are accounted for.

The remainder of the table shows how specific baseline characteristics vary with fund referral in more detail. The table shows that monthly rent averages \$705 among calls referred to assistance compared to \$457 for calls not referred. This gap is expected because referral depends on known, observable characteristics. For example, many delegate agencies only accept referrals to pay back rent, excluding callers requesting security deposits or utility payments who tend to owe lower rent. For similar reasons, those referred to funds have higher monthly income at the time of the call. However, the last columns shows that after controlling for Z_{it} this and most other differences are much smaller (closer to zero) than the raw differences. The difference in monthly rent reported by callers at the time of the call flips signs; those referred to funds have \$248 greater monthly rent but conditional on fund characteristics this difference becomes −\$20. Similarly, the difference in reported monthly earnings goes from \$260 to −\$57. These figures are based on information taken during calls to the HPCC, but we can verify this information by looking at pre-call earnings data from linked W-2s. Pre-call average earnings and AGI data are reported near the bottom of the table. These pre-period differences also become smaller after the controls in Z_{it} are included.

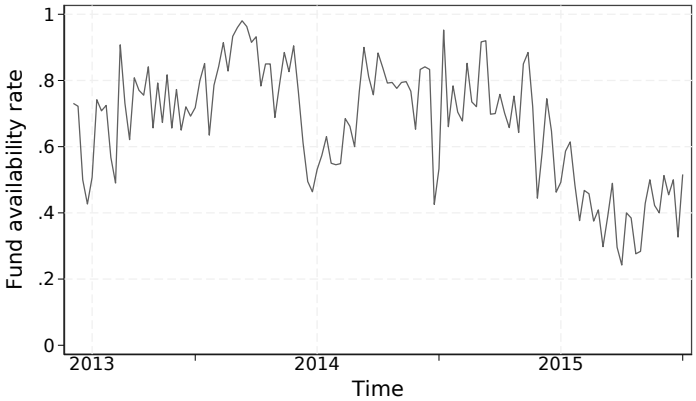
As indicated by the highly similar predicted wage and income results at the end of the table, the differences in column 4 have countervailing implications for how referred and non-referred households differ. With controls, referred individuals have slightly lower monthly income and rent, and are less likely to receive SNAP. But they report higher EITC participation and earnings.

to the HPCC, but that discussion does not permit this same analysis of the role of screening as (a) it does not include economic outcomes like employment & AGI (b) it does not include the sample used in Figure A.3 and (c) it is limited to a single year prior to the year of the call.

The version of Table A.7 for our sample of the lowest earners is reported in Table A.8; the higher-earners version is in Appendix Table A.9. Notably, we find stronger evidence that assistance increases earnings for low-earnings households. The tables show stronger balance across treatment and control in the low-wage group for several baseline characteristics, both before and after controlling for Z_{it} characteristics. For example, in the low-wage group pre-call adjusted gross income differs by a statistically insignificant \$925 in raw means and \$1,265 after adjusting for Z_{it}, about one-quarter and one-half of the respective gaps in the full sample. As in the full sample, any such differences tend to be offsetting and aggregate to negligible differences in predicted outcomes.

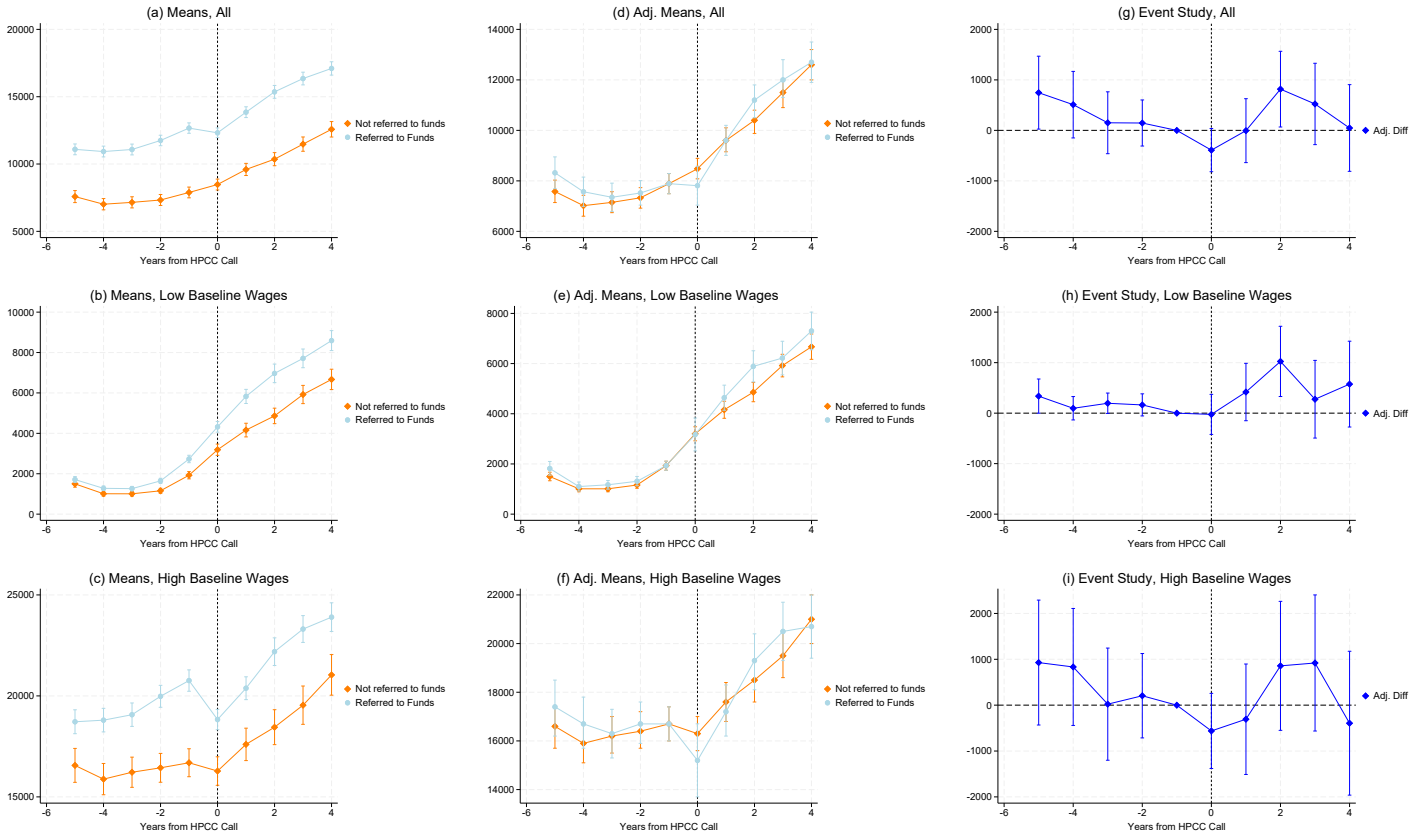
Tables and Figures

Fig. 1. Funding Rate by Week



Source: Homelessness Prevention Call Center data and authors' calculations.
Note: Figure shows availability of funding by week for the sample of all eligible callers between mid-2013 and the end of 2015.

Fig. 2. Wages, Trends in Means and Event Studies



Source: Homelessness Prevention Call Center data, IRS Form W-2 and authors' calculations. Note: All of these figures report wages from IRS Form W-2. The leftmost column of figures presents unadjusted mean wages for callers referred and not referred to funds by year relative to the year of the call. For the middle column of figures, for each year from 5 years before the call to 4 years after, we estimate equation (1). See notes of Table 3 for a full list of the control variables in this regression. We then plot the no-referral means from the leftmost column of figures as well as these means adjusted for the predicted effect of referral by adding our estimate of β_τ to the no-referral mean. The third column of figures plots the coefficients β_τ from each of these regressions. Release authorization CBDRB-FY23-0267 & CBDRB-FY25-0472.

Table 1: Sample Composition under Various Restrictions

	(1)	(2)	(3)	(4)	(5)
	Neighbors	Callers	Eligible Callers	Eligible Callers in Analysis Sample	Eligible Callers in Analysis Sample Low-wage
Age	43.42	40.16	38.41	38	38.82
	(0.0135)	(0.0598)	(0.0864)	(0.124)	(0.226)
Male	0.466	0.221	0.187	0.188	0.216
	(0.000356)	(0.00176)	(0.00265)	(0.00384)	(0.00646)
Citizen	0.796	0.960	0.975	0.972	0.974
	(0.000287)	(0.000831)	(0.00107)	(0.00163)	(0.00249)
Foreign-born	0.171	0.0336	0.0209	0.0248	0.0252
	(0.000269)	(0.000765)	(0.000972)	(0.00153)	(0.00246)
White	0.199	0.0618	0.0458	0.0433	0.0479
	(0.000285)	(0.00102)	(0.00142)	(0.00200)	(0.00336)
Black	0.521	0.729	0.764	0.770	0.749
	(0.000356)	(0.00189)	(0.00289)	(0.00413)	(0.00681)
Asian	0.0343	0.00274	0.00199	0.00202	0.000988
	(0.000130)	(0.000222)	(0.000303)	(0.000441)	(0.000494)
Hispanic	0.134	0.0956	0.0767	0.0795	0.0785
	(0.000243)	(0.00125)	(0.00181)	(0.00266)	(0.00423)
Other race	0.0152	0.0166	0.0176	0.0162	0.0156
	(8.72e-05)	(0.000543)	(0.000894)	(0.00124)	(0.00195)
Observations	1,966,000	55,500	21,500	10,500	4,000

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: Table reports simple mean demographic characteristics of callers from 2013-2015 in various samples and their neighbors (people living in the same set of Census blocks). The sample changes across columns. Column 1 (Neighbors) covers people living in the same Census block as any HPCC caller. Column 2 (Callers) covers all HPCC callers, with one observation for each year in which a call was made. Remaining columns are subsets of column 2. Column 3 (Eligible Callers) covers callers identified as meeting general eligibility at the time of at least one call. Column 4 (Eligible Callers in Analysis Sample) limits the analysis to callers with at least one call in our main analysis sample, which restricts to eligible callers with no prior call in the past 6 months. Column 5 (Eligible Callers in Analysis Sample, Low-wage) further limits the analysis to the sample of callers with below median pre-call wages. Age is in years. Other rows report the share of callers having the indicated demographic characteristic. Standard errors are in parentheses. Release authorization CBDRB-FY24-0033.

Table 2: Baseline Characteristics, by Referral Status

	Referred to Funds	Not Referred	Difference	Adjusted Difference	Adj. Diff as % of Mean
Monthly income (1000s)	1.37	1.11	0.26***	-0.057***	-5
			(0.016)	(0.017)	
Number of adults	1.39	1.45	-0.059***	-0.049**	-3
			(0.017)	(0.022)	
Number of minors	1.30	1.53	-0.23***	-0.16***	-10
			(0.035)	(0.046)	
Age	38.6	40.6	-1.95***	0.59*	1
			(0.30)	(0.34)	
Female	0.79	0.84	-0.052***	-0.028**	-3
			(0.0089)	(0.012)	
Hispanic	0.084	0.058	0.026***	0.018**	31
			(0.0058)	(0.0076)	
Black	0.87	0.92	-0.043***	-0.025***	-3
			(0.0069)	(0.0094)	
Other Non-White	0.035	0.026	0.0086**	0.010**	38
			(0.0039)	(0.0052)	
Reason for call: lost job	0.18	0.16	0.020**	0.011	7
			(0.0087)	(0.010)	
Reason for call: job cut hours	0.068	0.051	0.017***	-0.0069	-14
			(0.0054)	(0.0067)	
Reason for call: med. wrk absence	0.065	0.044	0.022***	0.0082	19
			(0.0051)	(0.0063)	
Reason for call: health	0.017	0.034	-0.018***	-0.0026	-8
			(0.0038)	(0.0048)	
Reason for call: housing change	0.096	0.19	-0.095***	0.016*	8
			(0.0084)	(0.0093)	
Reason for call: lost benefits	0.053	0.059	-0.0062	-0.00084	-1
			(0.0054)	(0.0063)	
Reason for call: other	0.045	0.090	-0.045***	-0.019***	-21
			(0.0061)	(0.0066)	
Reason for call: missing	0.47	0.37	0.10***	-0.0058***	-2
			(0.011)	(0.0020)	
Receiving SNAP	0.54	0.42	0.13***	-0.018*	-4
			(0.011)	(0.010)	
Receiving child support	0.024	0.019	0.0051	0.0017	9
			(0.0033)	(0.0045)	
Receiving EITC	0.31	0.16	0.15***	0.026***	16
			(0.0093)	(0.0097)	
Receiving SSI	0.064	0.071	-0.0069	-0.020**	-28
			(0.0059)	(0.0080)	
Receiving TANF	0.035	0.040	-0.0051	-0.0048	-12

			(0.0045)	(0.0065)	
Receiving UI	0.039	0.025	0.014***	0.0057	23
			(0.0040)	(0.0056)	
Lives with family or friends	0.12	0.14	-0.019**	-0.037***	-26
			(0.0079)	(0.010)	
Rents home	0.86	0.84	0.019**	0.037***	4
			(0.0083)	(0.010)	
Joint F-test of balance	p = 0.00				
Pre-call average wages	11510	7394	4112***	1416***	19
			(254.9)	(340.7)	
Pre-call average AGI	20290	16740	3551***	2042**	12
			(394.8)	(999.9)	
Pred. wages, 2 yrs post call	13320	10420	2900***	-20.6	0
			(151.4)	(157.2)	
Pred. AGI, 2 yrs post call	20260	18570	1687***	109	1
			(101.2)	(106.3)	

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: The sample consists of 5300 callers referred to funds and 2800 callers not referred to funds. Sample sizes are rounded in compliance with disclosure restrictions. The sample is restricted to eligible callers who have not called in the prior six months with non-missing characteristic information. The "Referred to Funds" column presents means for callers referred to funds; the "Not Referred" column presents means for callers not referred to funds. The "Difference" column reports the raw difference in means, and the "Adjusted Difference" column corresponds to an estimate of ψ in equation (4). The "Adj. Diff as % of Mean" column reports the adjusted difference as a percent of the control mean. The first panel shows results for baseline characteristics reported by callers to the HPCC and included in X_{i0}, the second panel reports lagged outcomes from federal tax data, and the third panel reports fitted values from a regression of outcomes on all baseline variables listed in Appendix Table A.7. The joint test of balance comes from an F-test of the null that all coefficients on X_{i0} are zero after a regression of referral to funds on X_{i0} and Z_{it}. A full list of X_{i0} and Z_{it} control variables are listed in the notes of Table 3. Release authorizations CBDRB-FY23-0267 and CBDRB-FY26-0144.

Table 3: Effects on Earnings

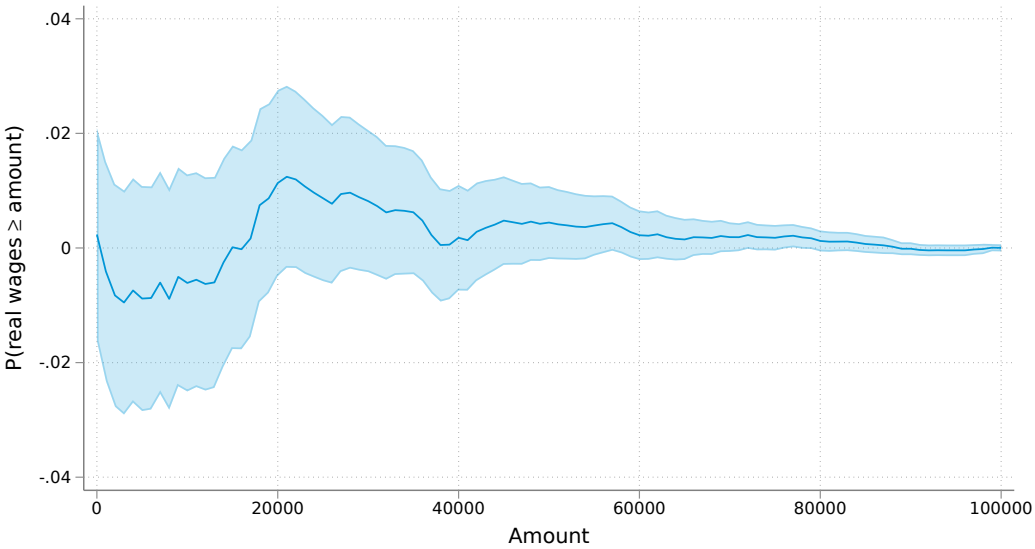
	No Lag	Main	No Xs	Fixed Effects	Zs × Qrtr	Fully Saturated	LASSO-Based
Full Sample	871**	197	156	574**	99.2	741	553***
	(358)	(281)	(288)	(252)	(286)	(559)	(130)
Controls × Quarter	No	No	No	No	Yes	No	Yes
Full Interactions	No	No	No	No	No	Yes	No
LASSO-selected Zs	No	No	No	No	No	No	Yes
N	40500	40500	40500	81500	40500	40500	41000
Low Wage	484*	431	427	1410***	431	483	494***
	(292)	(265)	(275)	(233)	(272)	(667)	(132)
Controls × Quarter	No	No	No	No	Yes	No	Yes
Full Interactions	No	No	No	No	No	Yes	No
LASSO-selected Zs	No	No	No	No	No	No	Yes
N	20000	20000	20000	40500	20000	20000	20000
High Wage	803	50	-112	311	-150	1540	514**
	(580)	(510)	(521)	(481)	(526)	(1270)	(223)
Controls × Quarter	No	No	No	No	Yes	No	Yes
Full Interactions	No	No	No	No	No	Yes	No
LASSO-selected Zs	No	No	No	No	No	No	Yes
N	20500	20500	20500	41000	20500	20500	21000

Note: Table reports effects of being referred to funds by the Homelessness Prevention Call Center on earnings for various samples and specifications. Mean earnings are 10400 for the full sample, 4940 for the low-wage group and 18100 for the high-wage group. The "No Lag" specification is estimated using equation (2) without the controls for $Y_{i,-1}$; the "Main" specification adds $Y_{i,-1}$; the "No Xs" specification includes Z_{it} and $Y_{i,-1}$ but not X_{i0}. The "Fixed Effects" specification is given by (3) in the text. "Zs x Qrtr" adds controls for interactions of each Z with a set of dummy variables for each year × quarter to the Main specification. The "Fully Saturated" specification further controls for dummies for all unique values of interactions of Z with each other then interacted with year × quarter dummies. The "LASSO-Based" column implements a double post-LASSO estimator that selects from among the controls from the fully saturated specification. Each of these specifications linearly controls for X. The fixed effects estimates use a sample from from five years prior through four years after each call; all other columns only use years zero to four. Standard errors clustered by person are in parentheses. Statistical significance at the 10, 5, and 1% levels are denoted by *, **, and ***, respectively.
The Z controls include: request type (i.e. rent assistance, security deposit, etc.), need amount categories, and indicators for military veterans, seniors, receipt of housing subsidies, owning a home, debt that exceeds one month of rent, income above twice the poverty line, and disability, as well as call characteristics including rank of the call within the day, the day of the week and month of the call, the time within the month (first five days, last five days, and middle days), a set of year-quarter interactions and interactions of need amount category with year-quarter. The X controls include: number of adults, number of minors, income, age and indicators for gender, crisis category, race, living situation, and income/benefit sources.
Release authorizations CBDRB-FY24-0033 & CBDRB-FY25-0472.

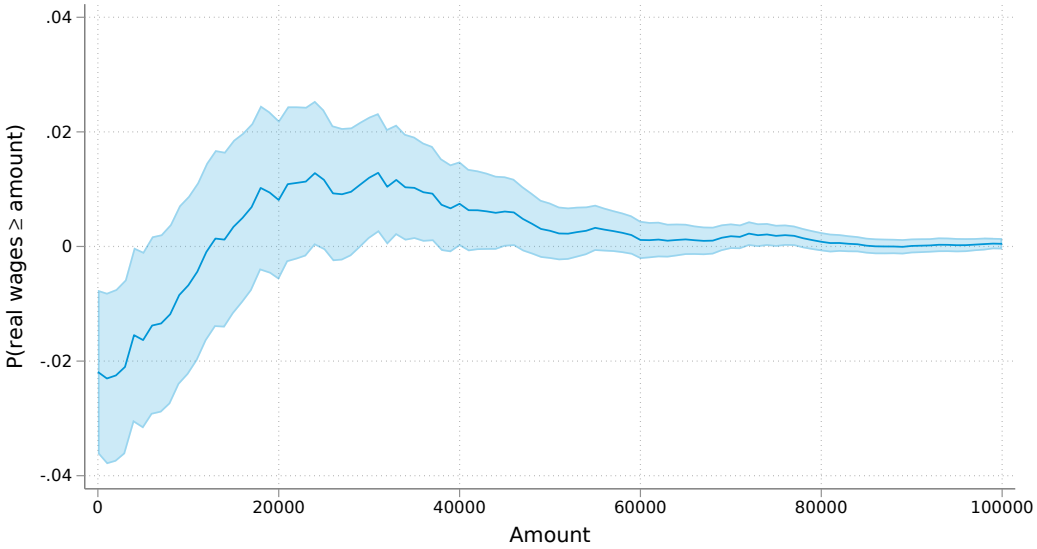
Appendix A. Additional Figures & Tables

Fig. A.1. Treatment Effects Across the Income Distribution

(a) Linear Controls



(b) Fixed Effects

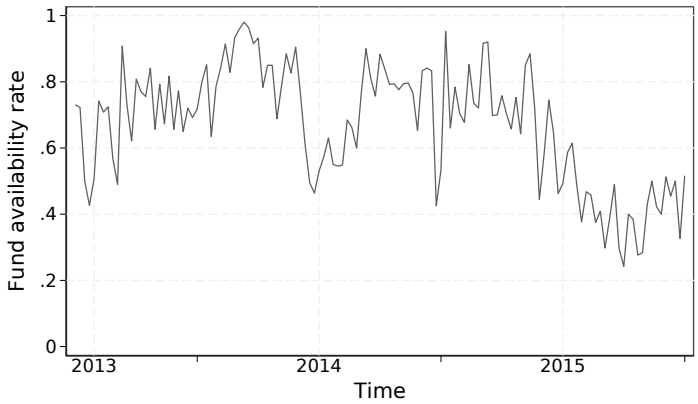


Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

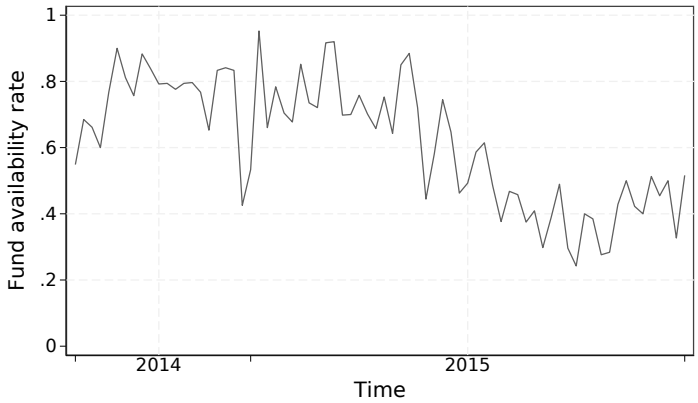
Note: Figures report estimates of the effect of being referred to funds by the Homelessness Prevention Call Center on the probability of having wage and salary income greater than various amounts between $0 and $100,000. Shaded regions represent 95 percent confidence intervals. Panel (a) is based off the specification in equation (2) and panel (b) is based off (3). Release authorization CBDRB-FY23-0267 and CBDRB-FY24-0033.

Fig. A.2. Funding Rate by Week

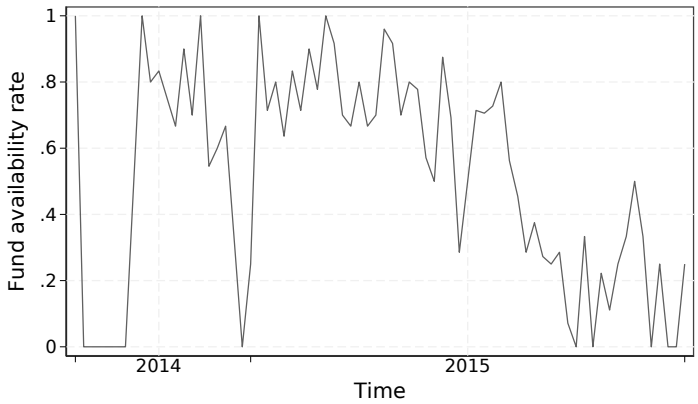
(a) Longer Sample - All



(b) Shorter Sample - All



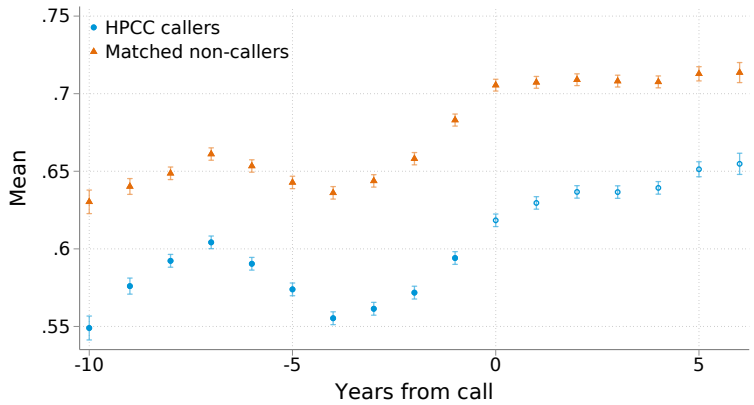
(c) Shorter Sample - Needs rent, need $\geq$ 1 month rent, need > \$900, no other housing subsidy



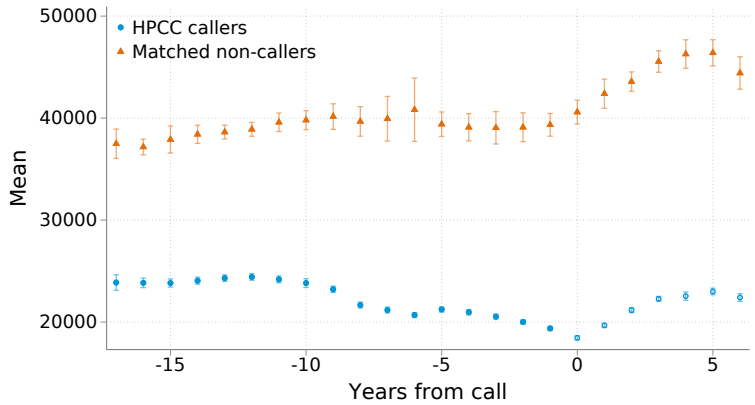
Source: Homelessness Prevention Call Center data and authors' calculations.
Note: Figure shows availability of funding by week for three samples of interest. Panel (a) is based on the sample of all eligible callers between mid-2013 and the end of 2015. Panel (b) is based on callers from August 2014 through December 2015, the period for which the data contain complete information on the reason for and amount of callers' needs. Panel (c) is based on the same period as panel (b) but limits the sample to callers with the characteristics indicated in the figure title.

Fig. A.3. Selection into Calling, All Callers

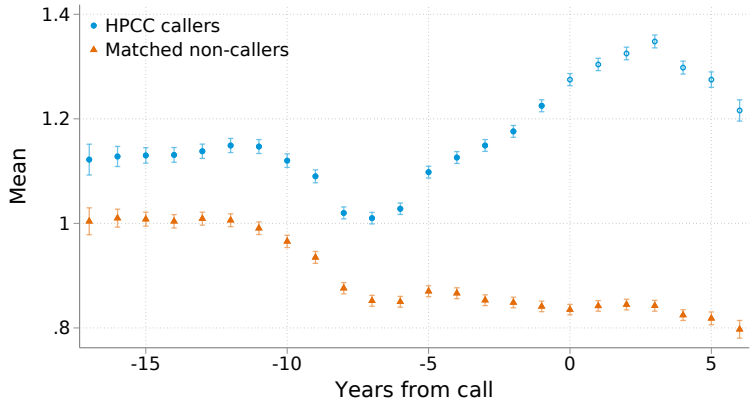
(a) Employment



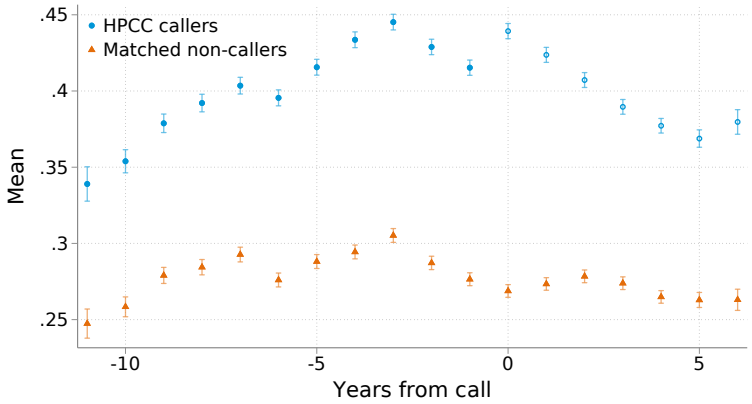
(b) Real AGI



(c) Number of children

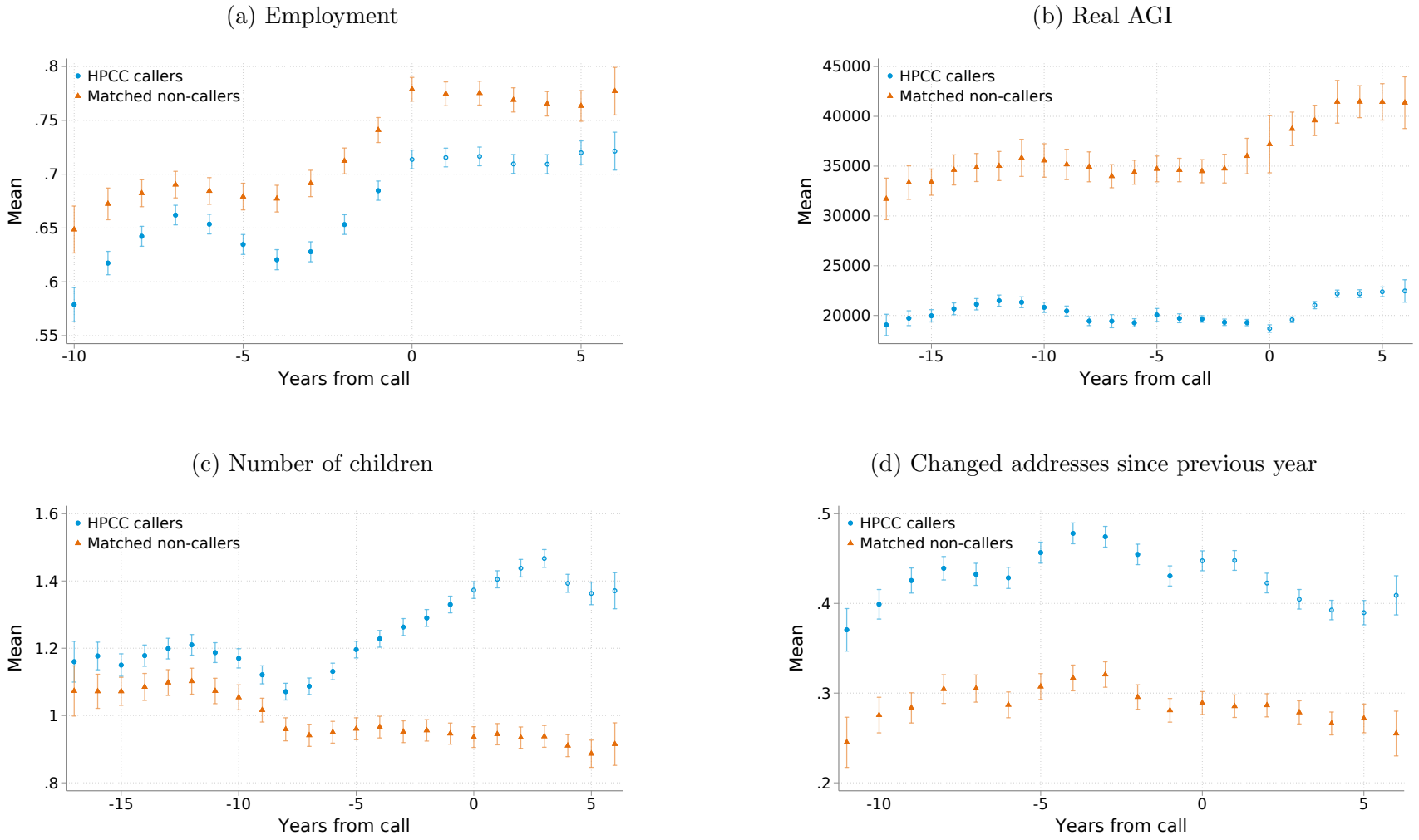


(d) Changed addresses since previous year



Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.
Note: Figure plots average outcomes for all callers to the Homelessness Prevention Call Center between mid-2013 and 2015 and matched non-callers. Callers are matched to non-callers based on demographic characteristics and place of residence in the year of the call, with matches drawn from the same Census block as callers where possible. Bars extending from plotted points represent 95 percent confidence intervals. Release authorization CBDRB-FY24-0033.

Fig. A.4. Selection into Calling, Full Analysis Sample



Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.
Note: Figure plots average outcomes for all callers in our main analysis sample and 2015 and matched non-callers. Callers are matched to non-callers based on demographic characteristics and place of residence in the year of the call, with matches drawn from the same Census block as callers where possible. Bars extending from plotted points represent 95 percent confidence intervals. Release authorization CBDRB-FY24-0033.

Table A.1: Lagged Y and No-Lagged-Y Specifications

	Lagged Outcome Control				No Lagged Outcome Control			
	Original	Interactions	Saturated	LASSO	Original	Interactions	Saturated	LASSO
Full Sample	197	99.2	741	553***	871**	664*	1490**	2440***
	(281)	(286)	(559)	(130)	(358)	(368)	(688)	(155)
Control Mean	10400	10400	10400	10400	10400	10400	10400	10400
N	40500	40500	40500	41000	40500	40500	40500	41000
Low Wages	431	431	483	494***	484*	480	637	769***
	(265)	(272)	(667)	(132)	(292)	(301)	(711)	(140)
Control Mean	4940	4940	4940	4940	4940	4940	4940	4940
N	20000	20000	20000	20000	20000	20000	20000	20000
High Wages	49.8	-150	1540	514**	803	495	2360*	2090***
	(510)	(526)	(1270)	(223)	(580)	(600)	(1410)	(244)
Control Mean	18100	18100	18100	18100	18100	18100	18100	18100
N	20500	20500	20500	21000	20500	20500	20500	21000

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, and authors' calculations.
Note: The table reports the effect of being referred to funds by the Homelessness Prevention Call Center on assorted outcomes for various samples and specifications. The four columns on the left are taken from Table 3 and include a control for wages in the year prior to calling the HPCC. All are based on equation (2) in the main text. The right 4 columns repeat this specification, but omit the lagged control. Statistical significance at the 10, 5, and 1% levels are denoted by *, **, and ***, respectively. Release authorization CBDRB-FY24-0033 & CBDRB-FY25-0472.

Table A.2: Varying Time Dummies

	Original	No Lag	Qtr	Qtr No Lag	Month	Month No Lag	Week	Week No Lag
Wages	197	871**	99.2	664*	96.2	967***	219	1010***
	(281)	(358)	(286)	(368)	(284)	(366)	(306)	(393)
Control Mean	10400	10400	10400	10400	10400	10400	10400	10400
N	40500	40500	40500	40500	40500	40500	40500	40500
Low Wages	431	484*	431	480	427	468	609*	769**
	(265)	(292)	(272)	(301)	(273)	(302)	(324)	(353)
Control Mean	4940	4940	4940	4940	4940	4940	4940	4940
N	20000	20000	20000	20000	20000	20000	20000	20000
High Wages	49.8	803	-150	495	-179	831	-35.4	890
	(510)	(580)	(526)	(600)	(535)	(612)	(596)	(678)
Control Mean	18100	18100	18100	18100	18100	18100	18100	18100
N	20500	20500	20500	20500	20500	20500	20500	20500

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, and authors' calculations.
Note: The table shows estimates of equation (2) where Z controls are interacted with quarter, month, or week-specific dummy variables as controls. Release authorizations CBDRB-FY24-0033 & CBDRB-FY25-0472.

Table A.3: Effect of Referral to Funds on Other Outcomes

	Ctrl Mean	Main Specification			Z × Qrtr Dummies			Post LASSO		
	Full Sample	Full Sample	Low Wage	High Wage	Full Sample	Low Wage	High Wage	Full Sample	Low Wage	High Wage
Employed	0.64	0.01	0.01	0.01	0.01	0.01	0.01	.021***	.013**	.014**
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)
Employed (alt)	0.43	0.000	0.003	-0.003	-0.002	0.004	-0.007	.0171***	.0112*	.0137**
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.00)	(0.01)	(0.01)
Tax Unit Wages	15660	344	609.6	240.3	190	468	-20.7	658***	665**	576*
		(376)	(444)	(582)	(398)	(446)	(641)	(201)	(266)	(296)
AGI	19130	714.2*	811.3**	692.9	419	540	289	760***	1340***	522*
		(369)	(396)	(579)	(367)	(380)	(612)	(192)	(263)	(295)
Filed 1040	0.62	0.012	.02198*	0.002	0.014	.0221*	0.005	.0116***	.0138**	-0.001
		(0.008)	(0.011)	(0.012)	(0.009)	(0.012)	(0.012)	(0.004)	(0.006)	(0.006)
1099-Misc	0.09	-0.002	-0.001	0.000	0.000	-0.002	0.009	.00813**	.00887**	0.002
		(0.006)	(0.008)	(0.010)	(0.006)	(0.008)	(0.010)	(0.003)	(0.004)	(0.005)
1099-G	0.41	0.015	.0225*	0.002	.0188*	.0235*	0.006	.0266***	.0261***	0.005
		(0.009)	(0.012)	(0.015)	(0.010)	(0.012)	(0.015)	(0.005)	(0.007)	(0.008)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, and authors' calculations.

Note: Other than the first column, which reports mean outcomes for non-referred callers, the table reports the effect of being referred to funds by the Homelessness Prevention Call Center on assorted outcomes for various samples and specifications all based on equation (2). The main specification includes controls for characteristics of callers, as well as a measure of earnings from the year before a person called. Statistical significance at the 10, 5, and 1% levels are denoted by *, **, and ***, respectively. Release authorization CBDRB-FY24-0033 & CBDRB-FY25-0472..

Table A.4: Heterogeneous Effects, Linear Controls

	Full Sample			Low Wage		
	Wages	AGI	Employed	Wages	AGI	Employed
Full Sample	197.4	714.2*	0.00757	430.6	811.3**	0.00734
	(280.8)	(369.2)	(0.00870)	(265.1)	(395.8)	(0.0128)
Lost Job	1257	2818***	-0.00894	1970**	1643	-0.00687
	(912)	(1094)	(0.0216)	(936.1)	(1338)	(0.0366)
Other Shock	-160.4	609.5	0.0174	163.5	631.9	0.0274
	(424.4)	(565.5)	(0.0143)	(398.6)	(531.9)	(0.0206)
Has Kids	63.49	505.1	0.00174	319.7	280.7	-0.00104
	(361.1)	(382)	(0.0106)	(354)	(342.6)	(0.0168)
No Kids	570.4	1733*	0.0179	758.8*	1705	0.0261
	(440.2)	(984.9)	(0.0149)	(400.7)	(1526)	(0.0198)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.
Note: Table reports effects of being referred to funds by the Homelessness Prevention Call Center on income and employment outcomes for various samples using the specification described in equation (2). Statistical significance at the 10, 5, and 1% levels are denoted by *, **, and ***, respectively. Release authorization CBDRB-FY24-0033.

Table A.5: Heterogeneous Effects, Zs × Qrtr Interaction Specifications

	Full Sample			Low Wage		
	Wages	AGI	Employed	Wages	AGI	Employed
Full Sample	99.2	419	0.00705	431	540	0.00754
	(286)	(367)	(0.00891)	(272)	(380)	(0.0131)
Lost Job	928	1760*	-0.0123	1850*	2130	-0.0291
	(938)	(1070)	(0.0224)	(1090)	(1430)	(0.0405)
Other Shock	-321	174	0.0203	126	412	0.0235
	(450)	(576)	(0.0149)	(431)	(586)	(0.0216)
Has Kids	124	399	0.000753	414	397	0.000213
	(363)	(387)	(0.0108)	(367)	(363)	(0.0172)
No Kids	310	764	0.0278*	549	-792	0.0285
	(452)	(1020)	(0.0153)	(417)	(1620)	(0.0202)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.
Note: Table reports effects of being referred to funds by the Homelessness Prevention Call Center on income and employment outcomes for various samples using the specification described in equation (2). Statistical significance at the 10, 5, and 1% levels are denoted by *, **, and ***, respectively. Release authorization CBDRB-FY24-0033 & CBDRB-FY25-0472.

Table A.6: Representativeness of the Analysis Sample

	Has PIK	No PIK	Dif.	Adj. Dif.
Number of adults	1.41	1.54	-0.13***	-0.093**
			(0.046)	(0.045)
Number of minors	1.38	1.44	-0.066	-0.11
			(0.075)	(0.073)
Monthly rent	618.2	705.4	-87.1***	-18.4*
			(18.7)	(11.0)
Net dollars owed	951.6	998.2	-46.6	28.6
			(31.8)	(20.3)
Monthly income (thousands)	1.28	1.44	-0.15***	-0.069**
			(0.037)	(0.030)
Veteran	0.016	0.025	-0.0088	6.2e-18**
			(0.0078)	(2.6e-18)
Senior	0.038	0.025	0.013	-3.3e-18
			(0.0080)	(2.1e-18)
Nbd median income	39480	36970	2511***	2377***
			(848.3)	(832.2)
Nbd pct Black	0.65	0.53	0.12***	0.12***
			(0.019)	(0.020)
Age	39.3	39.2	0.14	-0.034
			(0.64)	(0.57)
Reason missing	0.44	0.47	-0.033	-0.0070
			(0.025)	(0.0051)
Reason: lost job	0.17	0.18	-0.0062	-0.0098
			(0.020)	(0.017)
Reason: job cut hours	0.062	0.044	0.018*	0.015
			(0.011)	(0.011)
Reason: medical work absence	0.058	0.057	0.00081	0.0030
			(0.012)	(0.012)
Reason: health	0.023	0.022	0.00077	-0.00098
			(0.0075)	(0.0074)
Reason: housing change	0.13	0.10	0.028*	0.018
			(0.015)	(0.013)
Reason: lost benefits	0.055	0.049	0.0055	-0.00039
			(0.011)	(0.011)
Reason: other	0.060	0.074	-0.014	-0.018
			(0.013)	(0.013)
Female	0.81	0.81	-0.0037	-0.021
			(0.020)	(0.020)
Hispanic	0.075	0.21	-0.13***	-0.13***
			(0.020)	(0.020)
White	0.074	0.14	-0.064***	-0.060***
			(0.017)	(0.017)
Black	0.89	0.77	0.12***	0.11***
			(0.021)	(0.021)
Other	0.032	0.074	-0.043***	-0.040***
			(0.013)	(0.013)
Receives housing subsidy	0.33	0.20	0.12***	-2.6e-17
			(0.021)	(1.7e-17)
Receiving SNAP	0.50	0.40	0.10***	0.087***
			(0.025)	(0.020)
Receives income from disability	0.047	0.035	0.012	-2.4e-17***
			(0.0094)	(7.1e-18)
Receiving child support	0.022	0.030	-0.0074	-0.0056
			(0.0086)	(0.0087)
Receiving EITC	0.26	0.33	-0.075***	-0.030**
			(0.024)	(0.015)
Receiving disability	0.047	0.035	0.012	-2.4e-17***
			(0.0094)	(7.1e-18)
Receiving SSI	0.067	0.079	-0.012	-0.014
			(0.014)	(0.013)
Receiving TANF	0.037	0.012	0.024***	0.022***
			(0.0059)	(0.0063)
Receiving UI	0.034	0.025	0.0097	0.013
			(0.0080)	(0.0083)
Owns home	0.018	0.030	-0.012	-3.3e-18
			(0.0086)	(4.0e-18)
Lives with family or friends	0.13	0.13	0.00053	0.0011
			(0.017)	(0.016)
Rents home	0.86	0.84	0.011	-0.0011
			(0.018)	(0.016)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: The table compares callers who do and do not have a PIK (protected identification key) from the US Census Bureau. The sample includes about 8200 individuals with a PIK and about 400 individuals without a PIK. Sample sizes are rounded in compliance with disclosure restrictions. The first two columns present means for each group. The third column reports the difference in means, estimated via linear regression; *, **, and *** denote significance at the 10%, 5% and 1% levels. The last column reports this difference after controlling for a vector of caller characteristics Z_{it} related to fund eligibility. Robust standard errors in parentheses. Release authorization CBDRB-FY23-0267.

Table A.7: Baseline Characteristics, by Referral Status

	Referred to Funds	Not Referred	Difference	Adjusted Difference	Adj. Diff as % of Mean
Some Fund-Specific Criteria, Z_{it}					
Veteran	0.015	0.018	-0.0036		
			(0.0030)		
Senior	0.027	0.058	-0.031***		
			(0.0049)		
Receives housing subsidy	0.19	0.58	-0.40***		
			(0.011)		
Owns home	0.018	0.018	-0.000071		
			(0.0031)		
Receives inc. from disability	0.045	0.051	-0.0062		
			(0.0050)		
Baseline Balance Variables, X_{it}					
Monthly income (1000s)	1.37	1.11	0.26***	-0.057***	-5
			(0.016)	(0.017)	
Number of adults	1.39	1.45	-0.059***	-0.049**	-3
			(0.017)	(0.022)	
Number of minors	1.30	1.53	-0.23***	-0.16***	-10
			(0.035)	(0.046)	
Age	38.6	40.6	-1.95***	0.59*	1
			(0.30)	(0.34)	
Female	0.79	0.84	-0.052***	-0.028**	-3
			(0.0089)	(0.012)	
Hispanic	0.084	0.058	0.026***	0.018**	31
			(0.0058)	(0.0076)	
White	0.085	0.053	0.032***	0.012	23
			(0.0057)	(0.0080)	
Black	0.87	0.92	-0.043***	-0.025***	-3
			(0.0069)	(0.0094)	
Other Race	0.035	0.026	0.0086**	0.010**	38
			(0.0039)	(0.0052)	
Reason missing	0.47	0.37	0.10***	-0.0058***	-2
			(0.011)	(0.0020)	
Reason: lost job	0.18	0.16	0.020**	0.011	7
			(0.0087)	(0.010)	
Reason: job cut hours	0.068	0.051	0.017***	-0.0069	-14
			(0.0054)	(0.0067)	
Reason: med. wrk absence	0.065	0.044	0.022***	0.0082	19
			(0.0051)	(0.0063)	
Reason: health	0.017	0.034	-0.018***	-0.0026	-8
			(0.0038)	(0.0048)	
Reason: housing change	0.096	0.19	-0.095***	0.016*	8

			(0.0084)	(0.0093)	
Reason: lost benefits	0.053	0.059	-0.0062	-0.00084	-1
			(0.0054)	(0.0063)	
Reason: other	0.045	0.090	-0.045***	-0.019***	-21
			(0.0061)	(0.0066)	
Receiving SNAP	0.54	0.42	0.13***	-0.018*	-4
			(0.011)	(0.010)	
Receiving child support	0.024	0.019	0.0051	0.0017	9
			(0.0033)	(0.0045)	
Receiving EITC	0.31	0.16	0.15***	0.026***	16
			(0.0093)	(0.0097)	
Receiving SSI	0.064	0.071	-0.0069	-0.020**	-28
			(0.0059)	(0.0080)	
Receiving TANF	0.035	0.040	-0.0051	-0.0048	-12
			(0.0045)	(0.0065)	
Receiving UI	0.039	0.025	0.014***	0.0057	23
			(0.0040)	(0.0056)	
Lives with family or friends	0.12	0.14	-0.019**	-0.037***	-26
			(0.0079)	(0.010)	
Rents home	0.86	0.84	0.019**	0.037***	4
			(0.0083)	(0.010)	
Other Selected Variables					
Monthly rent	704.8	456.9	247.9***	-20.1***	-4
			(7.91)	(6.74)	
Net dollars owed	935.8	981.2	-45.4*	75.7***	8
			(25.1)	(23.3)	
Nbd median income	39760	38970	795.2***	1185***	3
			(297.3)	(410.1)	
Nbd pct Black	0.64	0.67	-0.021***	-0.021**	-3
			(0.0077)	(0.011)	
Lagged Outcomes					
Pre-call average wages	11510	7394	4112***	1416***	19
			(254.9)	(340.7)	
Pre-call average AGI	20290	16740	3551***	2042**	12
			(394.8)	(999.9)	
Predicted Outcomes					
Pred. wages, 2 yrs post call	13320	10420	2900***	-20.6	0
			(151.4)	(157.2)	
Pred. AGI, 2 yrs post call	20260	18570	1687***	109	1
			(101.2)	(106.3)	

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: The sample consists of 5300 callers referred to funds and 2800 callers not referred to funds. Sample sizes are rounded in compliance with disclosure restrictions. The sample is restricted to eligible callers who have not called in the prior six months with non-missing characteristic information. The first two columns present means for each group. The third column reports the difference in means, and the fourth column reports this difference after controlling for a vector of caller characteristics Z_{it} related to fund eligibility; robust standard errors in parentheses. The final column reports the adjusted difference as a percent of the control mean. Release authorization CBDRB-FY23-0267.

Table A.8: Baseline Characteristics for Low-Wage Group, by Fund Availability

	Referred to Funds	Not Referred	Dif.	Adj. Dif.
Some Fund-Specific Criteria, Z_{it}				
Veteran	0.015	0.021	-0.0066	
			(0.0043)	
Senior	0.048	0.082	-0.035***	
			(0.0080)	
Receives housing subsidy	0.25	0.62	-0.37***	
			(0.015)	
Owns home	0.011	0.015	-0.0040	
			(0.0037)	
Receives income from disability	0.079	0.072	0.0076	
			(0.0084)	
Baseline Balance Variables, X_{it}				
Monthy income (thousands)	1.22	0.98	0.24***	-0.053**
			(0.020)	(0.022)
Number of adults	1.38	1.44	-0.060***	-0.030
			(0.023)	(0.030)
Number of minors	1.37	1.52	-0.15***	-0.083
			(0.051)	(0.067)
Age	38.9	41.8	-2.95***	0.19
			(0.46)	(0.50)
Female	0.77	0.82	-0.055***	-0.034**
			(0.013)	(0.017)
Hispanic	0.079	0.049	0.030***	0.020**
			(0.0076)	(0.0096)
White	0.087	0.056	0.031***	0.0093
			(0.0080)	(0.011)
Black	0.87	0.91	-0.039***	-0.018
			(0.0096)	(0.013)
Other	0.033	0.026	0.0075	0.0071
			(0.0053)	(0.0068)
Reason missing	0.49	0.37	0.11***	-0.0031
			(0.016)	(0.0025)
Reason: lost job	0.14	0.13	0.015	0.012
			(0.011)	(0.013)
Reason: job cut hours	0.048	0.024	0.024***	0.011
			(0.0057)	(0.0074)
Reason: medical work absence	0.038	0.026	0.012**	0.00031
			(0.0055)	(0.0071)
Reason: health	0.021	0.050	-0.029***	-0.0035
			(0.0060)	(0.0075)
Reason: housing change	0.14	0.22	-0.080***	0.014
			(0.012)	(0.014)

Reason: lost benefits	0.063	0.068	-0.0046	-0.0036
			(0.0079)	(0.0088)
Reason: other	0.060	0.11	-0.049***	-0.027***
			(0.0090)	(0.0099)
Receiving SNAP	0.62	0.45	0.17***	-0.0041
			(0.016)	(0.012)
Receiving child support	0.026	0.018	0.0083*	0.0026
			(0.0046)	(0.0059)
Receiving EITC	0.26	0.11	0.15***	0.026**
			(0.012)	(0.013)
Receiving SSI	0.11	0.098	0.0076	-0.021*
			(0.0096)	(0.012)
Receiving TANF	0.064	0.052	0.013*	0.0061
			(0.0074)	(0.011)
Receiving UI	0.017	0.011	0.0060	-0.00023
			(0.0037)	(0.0054)
Lives with family or friends	0.16	0.15	0.0071	-0.038**
			(0.012)	(0.015)
Rents home	0.83	0.83	-0.0031	0.038**
			(0.012)	(0.015)
Other Selected Variables				
Monthly rent	623.5	389.4	234***	-10.8
			(10.5)	(8.60)
Net dollars owed	859.4	877.4	-18.0	64.8**
			(22.0)	(28.9)
Nbd median income	39450	38480	969**	1652***
			(406.9)	(546.9)
Nbd pct Black	0.65	0.68	-0.022**	-0.025*
			(0.011)	(0.014)
Lagged Outcomes				
Pre-call average wages	1726	1320	406.1***	210.7***
			(57.4)	(76.6)
Pre-call average AGI	14210	13290	925	1265
			(672.6)	(1678)
Predicted Outcomes				
Predicted wages, 2 years post call	10760	8176	2586***	57.5
			(200)	(208.6)
Predicted AGI, 2 years post call	18780	17390	1389***	126.1
			(131.9)	(136.2)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: The table compares callers who are and are not referred to funds with the sample of callers with below-median pre-call wages. The first two columns present means for each group. The third column reports the difference in means;*, **, and *** denote significance at the 10%, 5% and 1% levels. The last column reports this difference after controlling for a vector of caller characteristics Z_{it} related to fund eligibility; robust standard errors in parentheses. Release authorization CBDRB-FY23-0267.

Table A.9: Baseline Characteristics for High-Wage Group, by Fund Availability

	Referred to Funds	Not Referred	Dif.	Adj. Dif.
Some Fund-Specific Criteria, Z_{it}				
Veteran	0.015	0.014	0.00074	
			(0.0041)	
Senior	0.010	0.021	-0.011**	
			(0.0046)	
Receives housing subsidy	0.13	0.53	-0.40***	
			(0.016)	
Owns home	0.023	0.022	0.0014	
			(0.0051)	
Receives income from disability	0.016	0.020	-0.0036	
			(0.0047)	
Baseline Balance Variables, X_{it}				
Monthy income (thousands)	1.50	1.31	0.19***	-0.055**
			(0.024)	(0.026)
Number of adults	1.39	1.45	-0.062**	-0.056*
			(0.025)	(0.031)
Number of minors	1.24	1.54	-0.30***	-0.22***
			(0.049)	(0.063)
Age	38.4	38.8	-0.31	0.91*
			(0.38)	(0.48)
Female	0.81	0.87	-0.062***	-0.019
			(0.012)	(0.017)
Hispanic	0.087	0.070	0.017*	0.013
			(0.0092)	(0.012)
White	0.084	0.049	0.035***	0.014
			(0.0081)	(0.012)
Black	0.87	0.92	-0.048***	-0.035**
			(0.010)	(0.014)
Other	0.035	0.026	0.0093	0.016**
			(0.0058)	(0.0080)
Reason missing	0.47	0.36	0.10***	-0.010***
			(0.017)	(0.0031)
Reason: lost job	0.21	0.21	0.0049	0.016
			(0.014)	(0.016)
Reason: job cut hours	0.084	0.091	-0.0068	-0.028**
			(0.0099)	(0.012)
Reason: medical work absence	0.087	0.069	0.018**	0.020*
			(0.0091)	(0.011)
Reason: health	0.014	0.011	0.0023	-0.0021
			(0.0038)	(0.0054)
Reason: housing change	0.063	0.15	-0.089***	0.011
			(0.011)	(0.012)

Reason: lost benefits	0.044	0.046	-0.0018	0.0010
			(0.0073)	(0.0092)
Reason: other	0.032	0.061	-0.028***	-0.0083
			(0.0078)	(0.0086)
Receiving SNAP	0.48	0.36	0.11***	-0.026
			(0.017)	(0.017)
Receiving child support	0.022	0.021	0.0016	-0.00024
			(0.0050)	(0.0070)
Receiving EITC	0.35	0.23	0.12***	0.018
			(0.015)	(0.014)
Receiving SSI	0.031	0.032	-0.0012	-0.0071
			(0.0061)	(0.0089)
Receiving TANF	0.011	0.023	-0.012**	-0.014**
			(0.0048)	(0.0067)
Receiving UI	0.057	0.046	0.011	0.0033
			(0.0075)	(0.011)
Lives with family or friends	0.086	0.12	-0.029***	-0.024*
			(0.011)	(0.014)
Rents home	0.89	0.86	0.028**	0.024*
			(0.012)	(0.014)
Other Selected Variables				
Monthly rent	771.4	556.4	215***	-25.5**
			(11.8)	(10.4)
Net dollars owed	997.9	1133	-135.4**	74.9
			(52.8)	(47.9)
Nbd median income	40010	39680	334.3	637.4
			(447.7)	(629.1)
Nbd pct Black	0.64	0.65	-0.013	-0.021
			(0.012)	(0.016)
Lagged Outcomes				
Pre-call average wages	19470	16360	3113***	1582***
			(373.2)	(519.9)
Pre-vall average AGI	24250	20540	3708***	1910***
			(466.8)	(654.6)
Predicted Outcomes				
Predicted wages, 2 years post call	15420	13700	1715***	-181.6
			(200.5)	(222.4)
Predicted wages, 2 years post call	21470	20300	1174***	67.0
			(144.3)	(159.6)

Source: Homelessness Prevention Call Center data, American Community Survey, Decennial Census, IRS Form 1040, Form W-2, IRS information returns, Census Numident, Master Address File Auxiliary Reference File, and authors' calculations.

Note: The table compares callers who are and are not referred to funds with the sample of callers with above-median pre-call wages. The first two columns present means for each group. The third column reports the difference in means;*, **, and *** denote significance at the 10%, 5% and 1% levels. The last column reports this difference after controlling for a vector of caller characteristics $Z_i t$ related to fund eligibility; robust standard errors in parentheses. Release authorization CBDRB-FY23-0267.